

UNIT-II
DATA CLEANING AND PREPROCESSING

Agenda



- Introduction
- Data Collection and Data Preprocessing
  - Data Collection Strategies
- Data Preprocessing Overview
  - Data Cleaning
- Data Integration and Transformation
  - Data reduction
  - Data discretization



Session-1

DATA COLLECTION AND STRATEGIES

DATA COLLECTION

What is Data?

- Fact and statistics collected together for reference or analysis
- Research data is defined as recorded factual material commonly retained by and accepted in the scientific community as necessary to validate research findings.

Data Collection and Types



- The process of **gathering and analyzing accurate data** from various sources to find answers to research problems, trends probabilities, etc., to evaluate possible outcomes is Known as Data Collection.
  - **Knowledge is power**
  - **Information is knowledge**
  - **Data is information in digitized form**
- An essential phase in all types of **research, analysis, and decision-making**, including that done in the **social sciences, business, and healthcare**.
- Analyzing the following questions before data collection
  - **What's the goal or purpose of this research?**
  - **What kinds of data are they planning on gathering?**
  - **What methods and procedures will collect, store, and process the information?**

Data Collection and Types



Quantitative Data Collection Methods



- Quantitative data collection refers to the **collection of numerical data** that can be analyzed using statistical methods.
- This type of data collection is often used in **surveys, experiments**, and other research methods.
- **Discrete and continuous** are the two major categories of quantitative data.

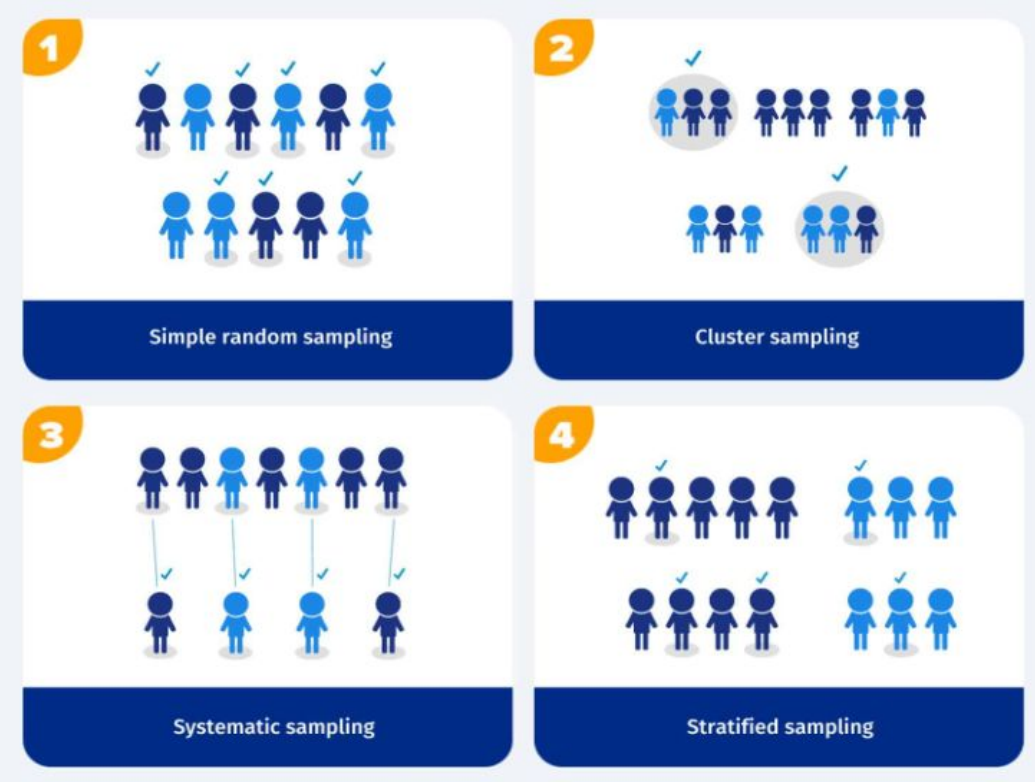
Example:

- If research is conducted to **find out the number of vehicles** parked inside the campus, then we get a whole number, which is an excellent example of **discrete data**.
- When research is limited to the study of **physical measurements** of the **population** like **height, weight, age,** then the result is an excellent example of continuous data.

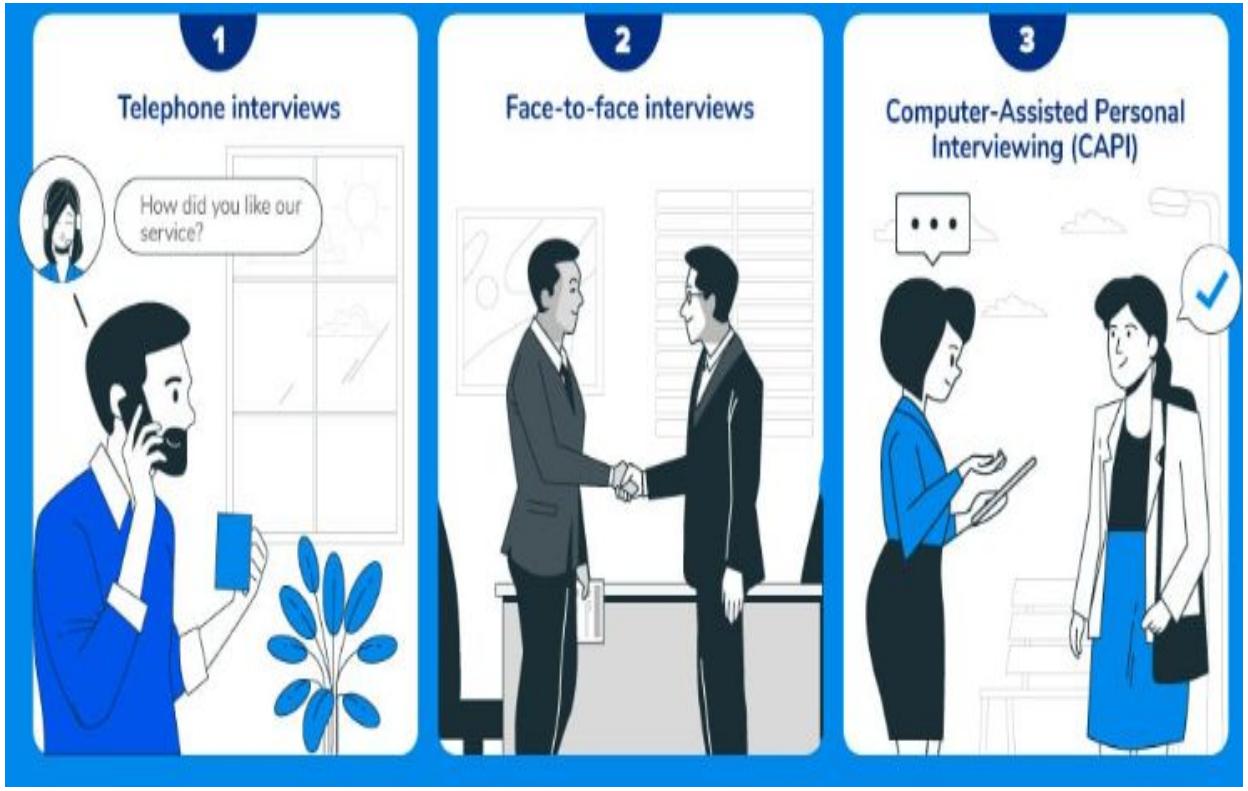
Quantitative Data Collection Methods



Quantitative Data Collection Methods



Probability Sampling



Interviews

Quantitative Data Collection Methods

- In a web-based questionnaire, the receive an email containing the survey link, clicking on which takes the respondent to a secure online survey tool from where he/she can take the survey or fill in the survey questionnaire.
- In a mail questionnaire, the survey is mailed out to a host of the sample population, enabling the researcher to connect with a wide range of audiences.
- The mail questionnaire typically consists of a packet containing a cover sheet that introduces the audience about the type of research and reason why it is being conducted along with a prepaid return to collect data online.



Surveys

Qualitative Data Collection Methods



Primary and Secondary Data Collection Methods

Primary Data Collection: Collection of original data directly from the source or through direct interaction with the respondents.

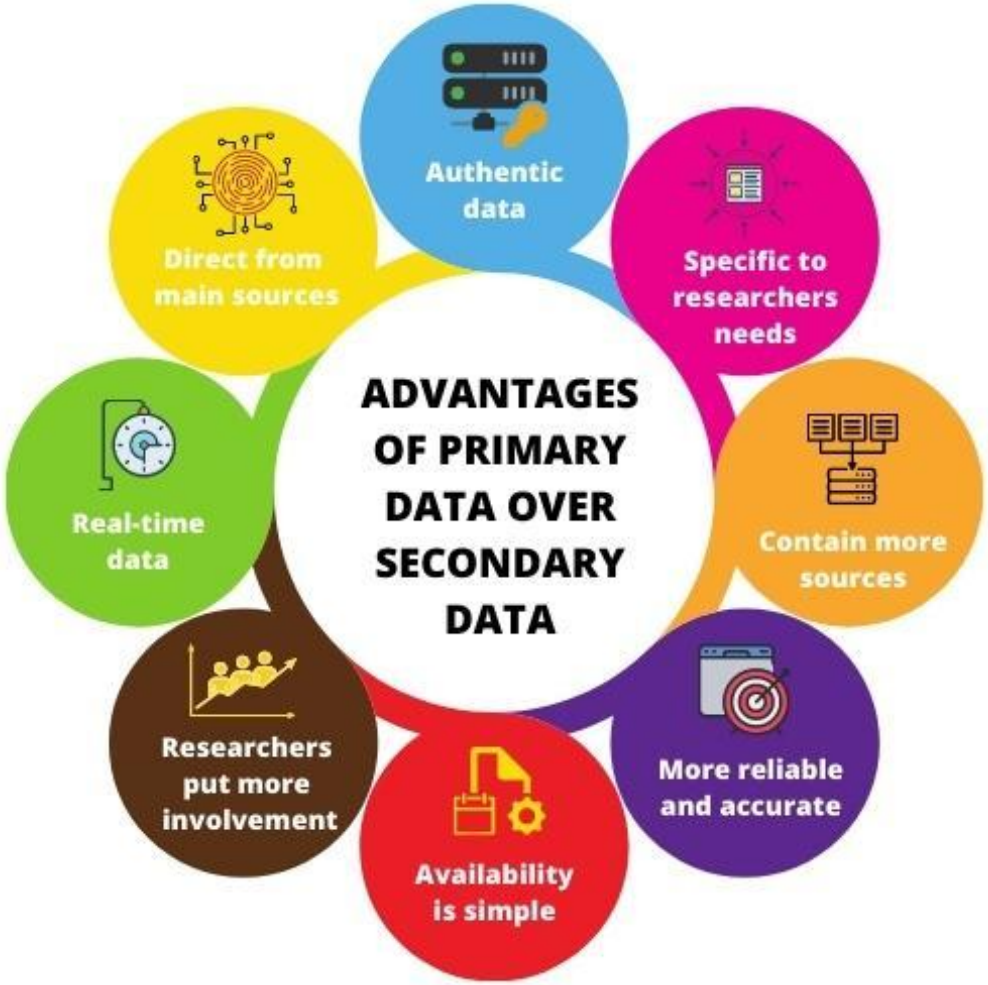
- Surveys and Questionnaires
- Interviews
- Observations
- Experiments
- Focus Groups

Secondary Data collection: Using existing data collected by someone else for a purpose different from the original intent.

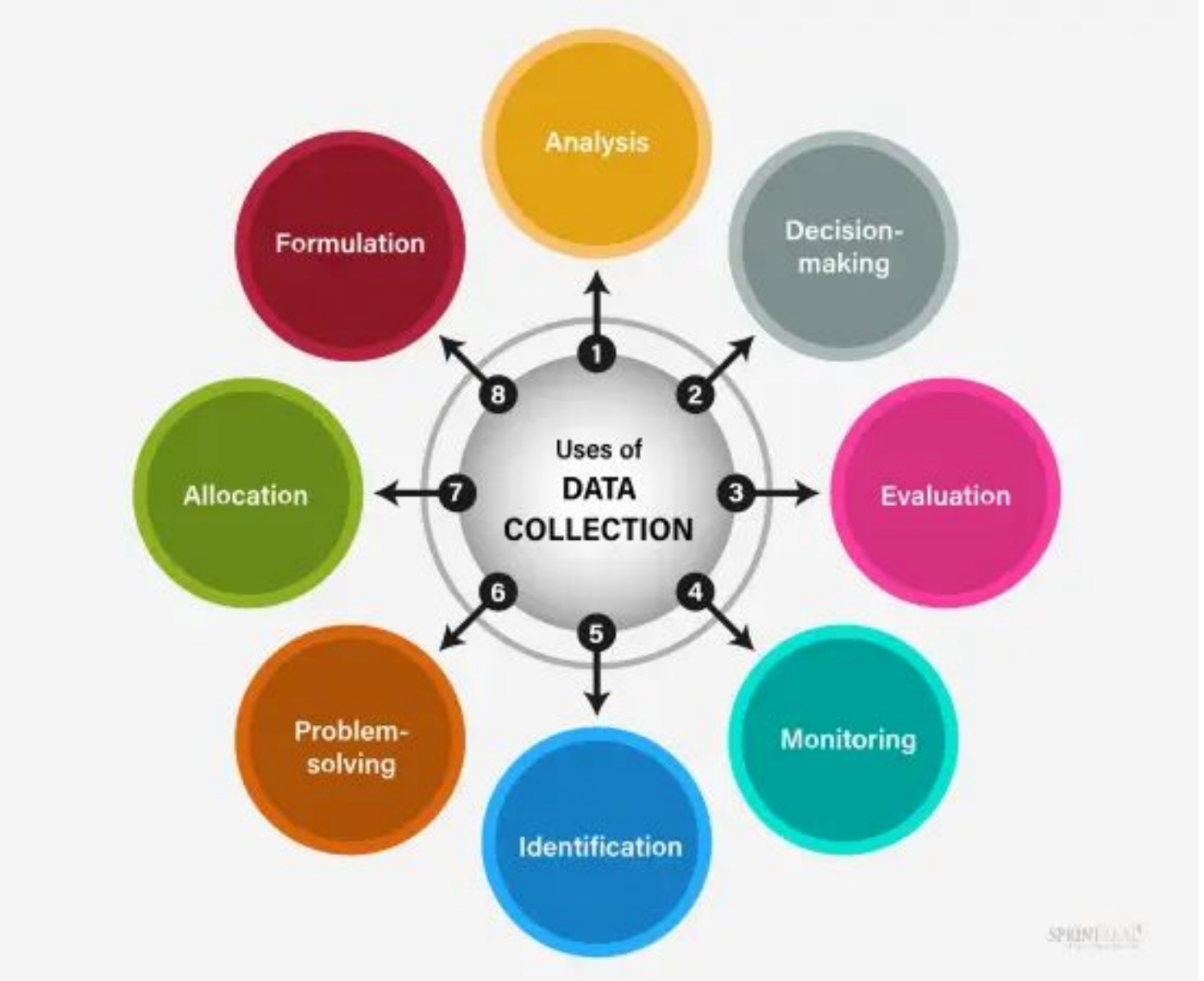
- Published Sources
- Online Databases
- Government and Institutional Records
- Publicly Available Data
- Past Research Studies



Advantages over Secondary Data Collection Methods



Use of Data Collection Process



Key Steps in Data Collection Process



- Decide What Data You Want to Gather
- Establish a Deadline for Data Collection
- Select a Data Collection Approach
- Gather Information
- Examine the Information and Apply Your Findings

Data Collection Process – some videos



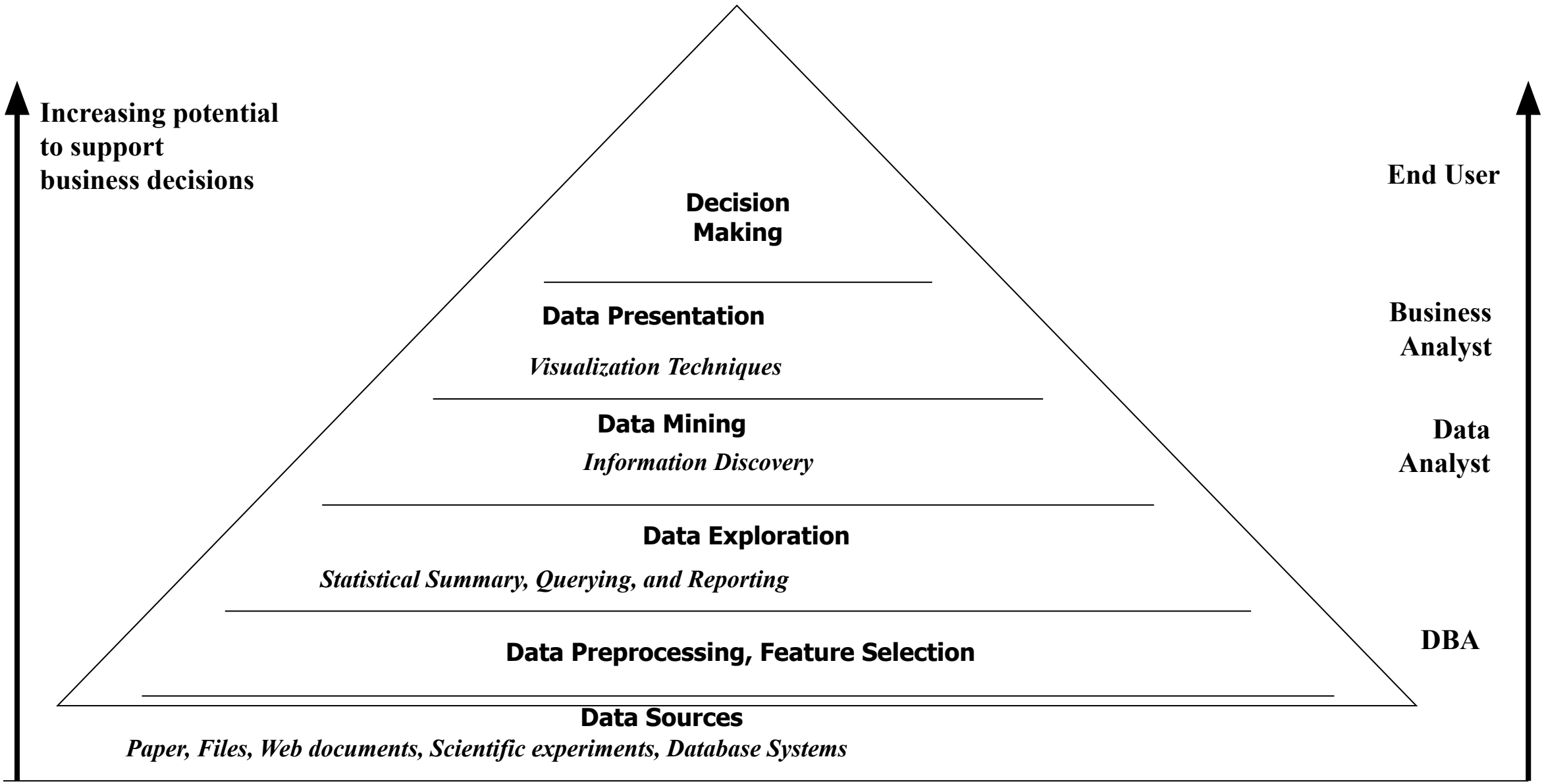
- https://www.youtube.com/watch?v=Lb6Gi6IR-Kc
- https://www.youtube.com/watch?v=Coe0N2xb8kk
- https://www.youtube.com/watch?v=_zyN3Wa_a7c



Session -II

DATA PREPROCESSING

How Data is Used



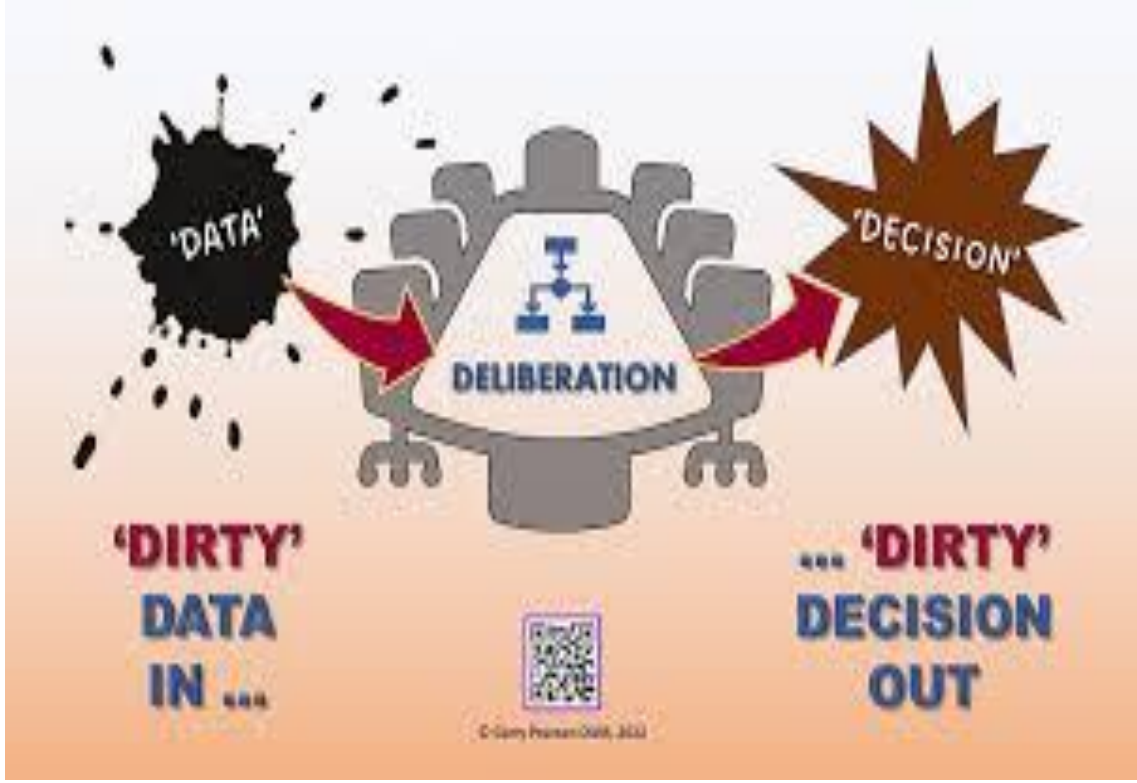
Why Data Preprocessing?

- Data in the real world is dirty
  - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
    - e.g., occupation=“ ”
  - noisy: containing errors or outliers
    - e.g., Salary=“-10”
  - inconsistent: containing discrepancies in codes or names
    - e.g., Age=“42” Birthday=“03/07/1997”
    - e.g., Was rating “1,2,3”, now rating “A, B, C”
    - e.g., discrepancy between duplicate records



Why Is Data Dirty?

- Incomplete data may come from
  - “Not applicable” data value when collected
  - Different considerations between the time when the data was collected and when it is analyzed.
  - Human/hardware/software problems
- Noisy data (incorrect values) may come from
  - Faulty data collection instruments
  - Human or computer error at data entry
  - Errors in data transmission
- Inconsistent data may come from
  - Different data sources
  - Functional dependency violation (e.g., modify some linked data)
- Duplicate records also need data cleaning



Data Pre-processing



Data preprocessing, a component of data preparation, describes any type of processing performed on raw data to prepare it for another data preprocessing procedure.

Data Pre-processing methods

- **Sampling,** which selects a representative subset from a large population of data;
- **Transformation**, which manipulates raw data to produce a single input;
- **Denoising**, which removes noise from data;
- **Imputation**, which synthesizes statistically relevant data for missing values;
- **Normalization**, which organizes data for more efficient access; and
- **Feature extraction**, which pulls out a relevant feature subset that is significant in a particular context.

Why is Data Pre-processing important?



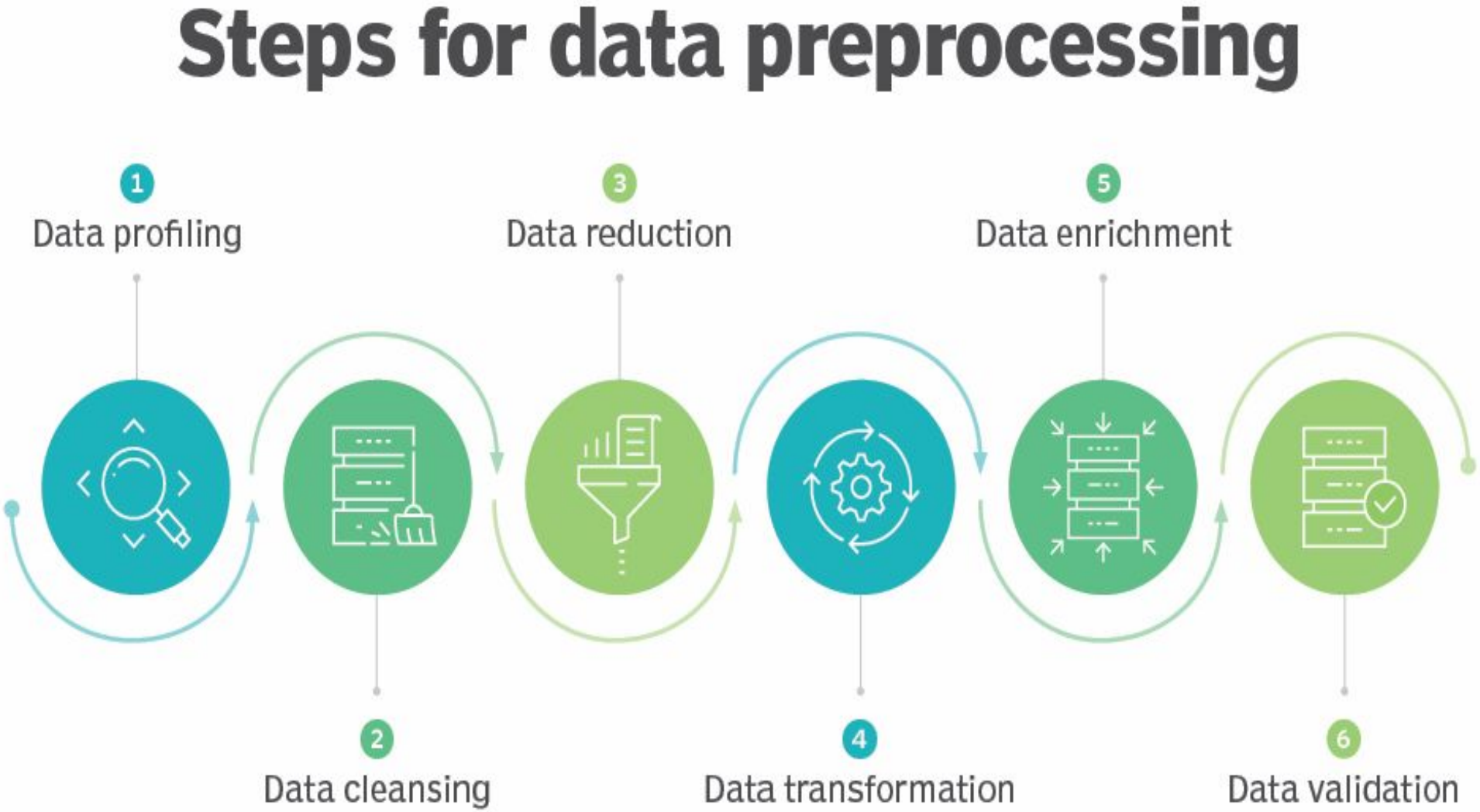
- Data science or AI development requires some type of data preprocessing **to provide reliable, precise and robust results** for enterprise applications.
- **Real-world data is messy** and is often created, processed and stored by a variety of humans, business processes and applications.
- Data set may be **missing individual fields**, **contain manual input errors**, or **have duplicate data** or **different names** to describe the same thing.
- Humans can often identify and rectify these problems in the data they use in the line of business, but data used **to train machine learning or deep learning algorithms** needs to be **automatically preprocessed.**

Data Pre Processing – some videos



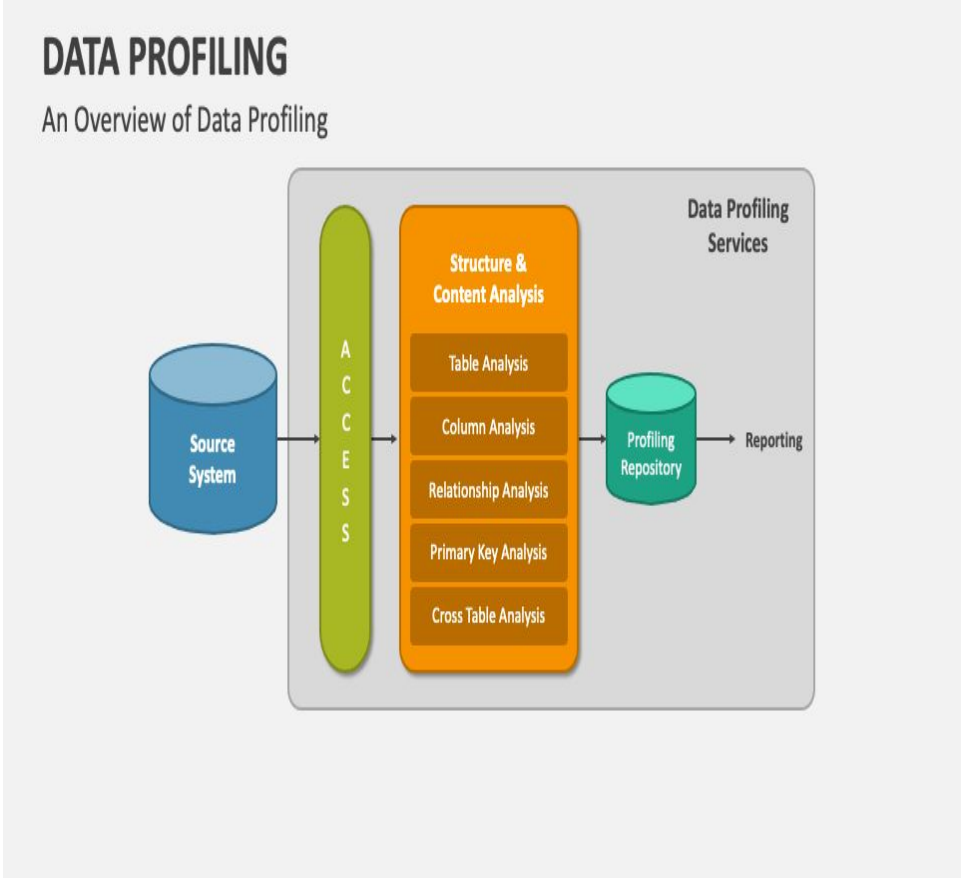
- https://www.youtube.com/watch?v=yAIEC7JUOMQ
- https://www.youtube.com/watch?v=bN5aZLp0gvw

Key steps in Data Pre-processing



Data Profiling

- **Data profiling** is the process of examining and analyzing data to understand its structure, content, quality, and relationships.
- This process is crucial in various stages of data management, including data cleansing, data integration, and data migration.
- By profiling data, you can detect anomalies, validate assumptions, and ensure that the data is fit for its intended use.



Steps in Data Profiling

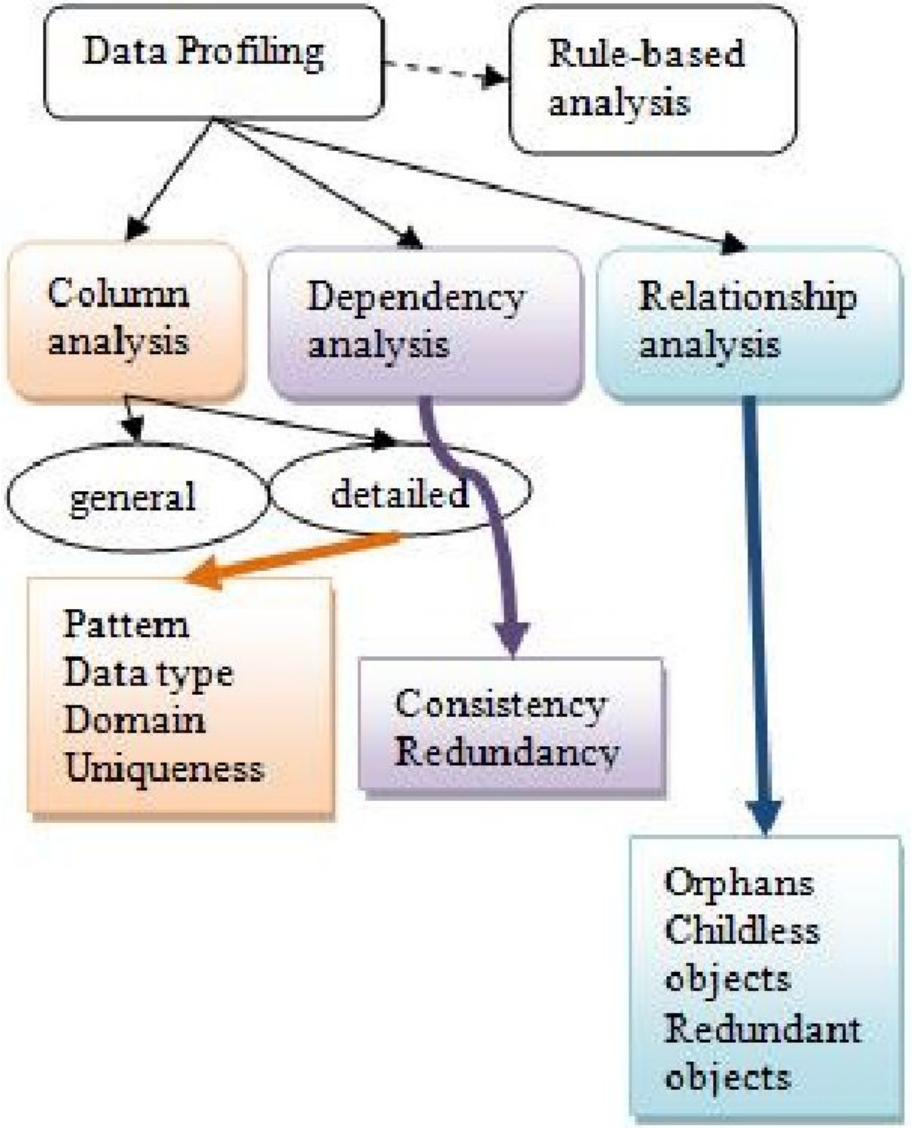
- **Structure Discovery -** is the process of examining and analyzing data to understand its structure, content, quality, and relationships.
- **Content discovery** — Content discovery focuses on data quality. Data needs to be processed for formatting and standardization, and then properly integrated with existing data in a timely and efficient manner.
- **Relationship discovery** — Relationship discovery identifies connections between different datasets.



Steps in Structure Discovery

Structure Discovery –

- is the process of examining and analyzing data to understand its structure, content, quality, and relationships. examines how your data is formatted and organized.
- It's like taking a census of your data sets, identifying the number and type of fields within each dataset, along with their content.
- This process ensures format consistency – for example, a phone number field shouldn't contain text or an uneven number of digits.



Example of Structure Discovery

	A	B	C	D	E
1	CustomerID	Name	Age	Email	Country
2	1	John Doe	28	john.doe@example.com	USA
3	2	Jane Smith	35	jane.smith@example.co	UK
4	3	Mary Johnson	22	mary.j@sample.com	USA
5	4	Chris Evans	45	chris.evans@domain.net	NULL
6	5	NULL	29	tom.clark@sample.org	Canada
7	6	Anna Brown	NULL	anna.b@example.com	USA
8	7	John Doe	28	john.doe@example.com	USA
9	8	Tom Clark	31	tom.clark@sample.org	Canada

Structure Discovery –

- is the process of examining and analyzing data to understand its structure, content, quality, and relationships. examines how your data is formatted and organized.
- It's like taking a census of your data sets, identifying the number and type of fields within each dataset, along with their content.
- This process ensures format consistency – for example, a phone number field shouldn't contain text or an uneven number of digits.

Steps in Content Discovery

Content Discovery –

- Content discovery involves examining each database's individual fields and elements to check the contents and quality.
- It involves looking into individual data records to identify errors and determine which rows in a dataset contain problems or systemic issues.

Example of Content Discovery

	A	B	C	D	E
1	CustomerID	Name	Age	Email	Country
2	1	John Doe	28	john.doe@example.com	USA
3	2	Jane Smith	35	jane.smith@example.co	UK
4	3	Mary Johnson	22	mary.j@sample.com	USA
5	4	Chris Evans	45	chris.evans@domain.net	NULL
6	5	NULL	29	tom.clark@sample.org	Canada
7	6	Anna Brown	NULL	anna.b@example.com	USA
8	7	John Doe	28	john.doe@example.com	USA
9	8	Tom Clark	31	tom.clark@sample.org	Canada

Here's a closer look at content profiling:

- Statistical summaries: Analyze minimum/maximum values for numerical fields and the frequency of values in categorical fields. This provides insights into the data range and quality.
- Null values and uniqueness: Check the number of null values, blanks, and unique values to understand data completeness, quality, and a field's relevance.
- Systemic Errors: Look for systemic errors like misspellings or inconsistent value representations ("Doctor" vs. "Dr."), which can disrupt analysis.

Steps in Relationship Discovery

Relationship Discovery –

- Unlike structure and content discovery, which focus on individual datasets, relationship discovery explores how your data connects across different sources.
- Discovering how parts of the data are interrelated. For example, key relationships between database tables, references between cells or tables in a spreadsheet.
- Understanding relationships is crucial to reusing data; related data sources should be united into one or imported in a way that preserves important relationships.

Example of Relationship Discovery

	A	B	C	D	E
1	CustomerID	Name	Age	Email	Country
2	1	John Doe	28	john.doe@example.com	USA
3	2	Jane Smith	35	jane.smith@example.co	UK
4	3	Mary Johnson	22	mary.j@sample.com	USA
5	4	Chris Evans	45	chris.evans@domain.net	NULL
6	5	NULL	29	tom.clark@sample.org	Canada
7	6	Anna Brown	NULL	anna.b@example.com	USA
8	7	John Doe	28	john.doe@example.com	USA
9	8	Tom Clark	31	tom.clark@sample.org	Canada

Here's a closer look at Relationship profiling:

- Uniqueness: Ensure that certain columns, like "CustomerID" or "Email," are unique identifiers. CustomerID" is unique except for the possible duplication in the example. "Email" should ideally be unique, but "tom.clark@sample.org" is associated with two different customers.
- Dependency: Check if values in one column depend on values in another (e.g., "Age" should not be NULL if "Name" is present).

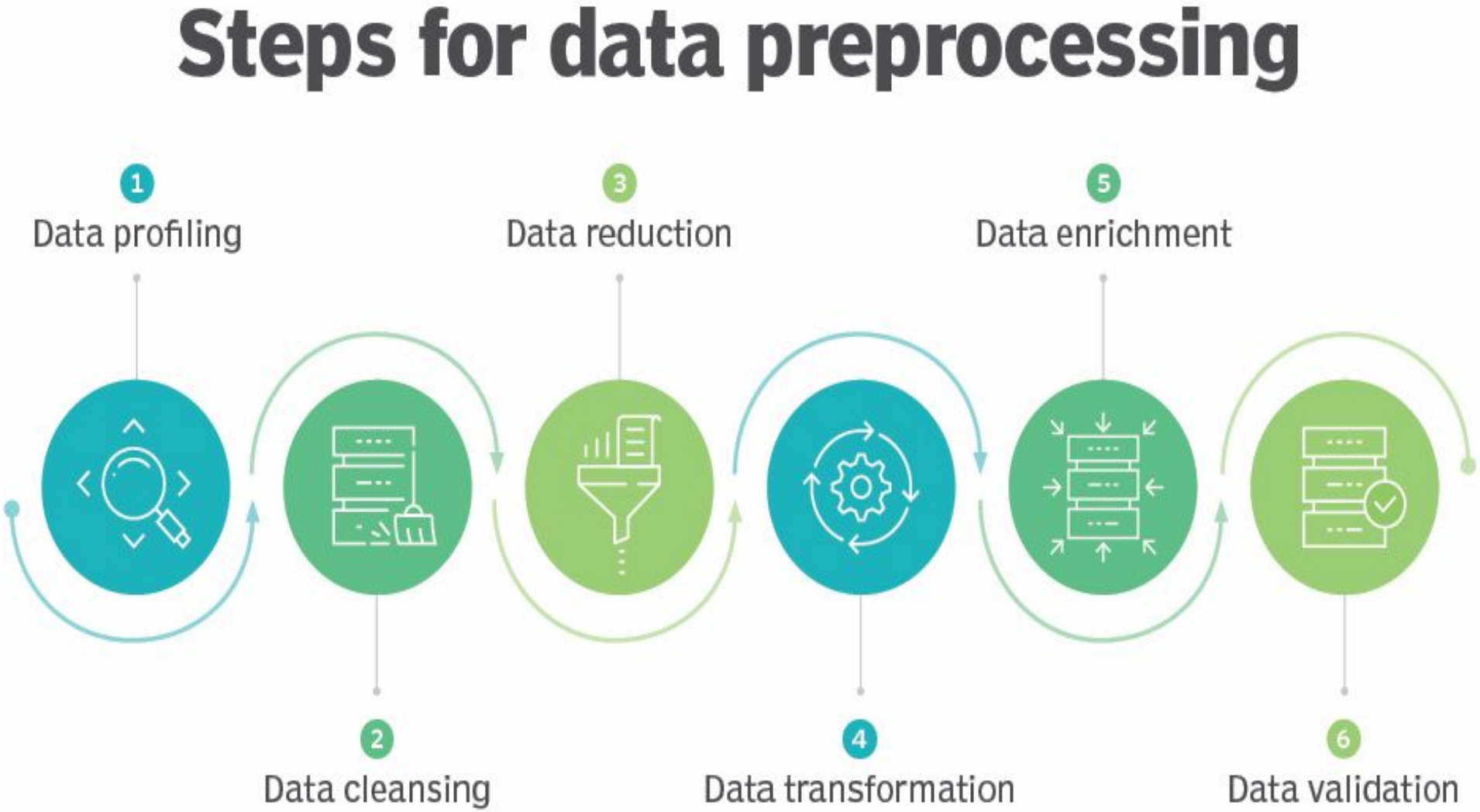
Data Profiling– some videos



- https://www.youtube.com/watch?v=HtaYjVwW-Mo

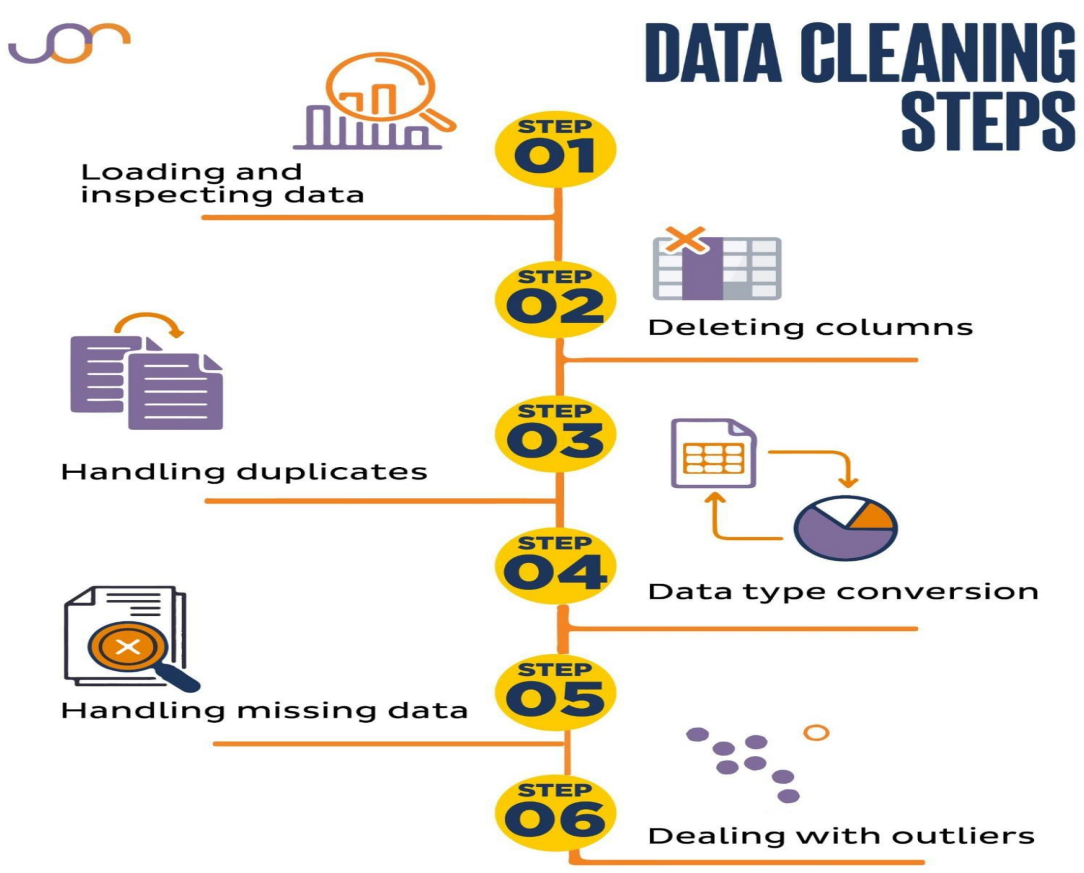
- https://www.youtube.com/watch?v=KZ7xSpLYD-o

Key steps in Data Pre-processing



What is Data Cleansing?

- Data cleansing or data scrubbing is the act of detecting and correcting(or removing)corrupt or inaccurate records from a record set, table, or database.
- Used mainly in databases, the term refers to identifying incomplete, incorrect, inaccurate, irrelevant, etc. parts of the data and then **replacing, modifying, or deleting the dirty data.**
- Data cleansing can occur within a single set of records, or between multiple sets of data that need to be merged, or which will work together.
- **Typos and spelling errors** are corrected, **mislabeled data** is properly labeled and files and **incomplete or missing entries** are completed.



Why is Data Cleaning Important?



- Data cleansing is a crucial step in the data preparation process, playing an important role in ensuring the accuracy, reliability, and overall quality of a dataset.
- Without proper data cleaning, inaccuracies, outliers, missing values, and inconsistencies can compromise the validity of analytical results.
- clean data facilitates more effective modeling and pattern recognition, as algorithms perform optimally when fed high-quality, error-free input.

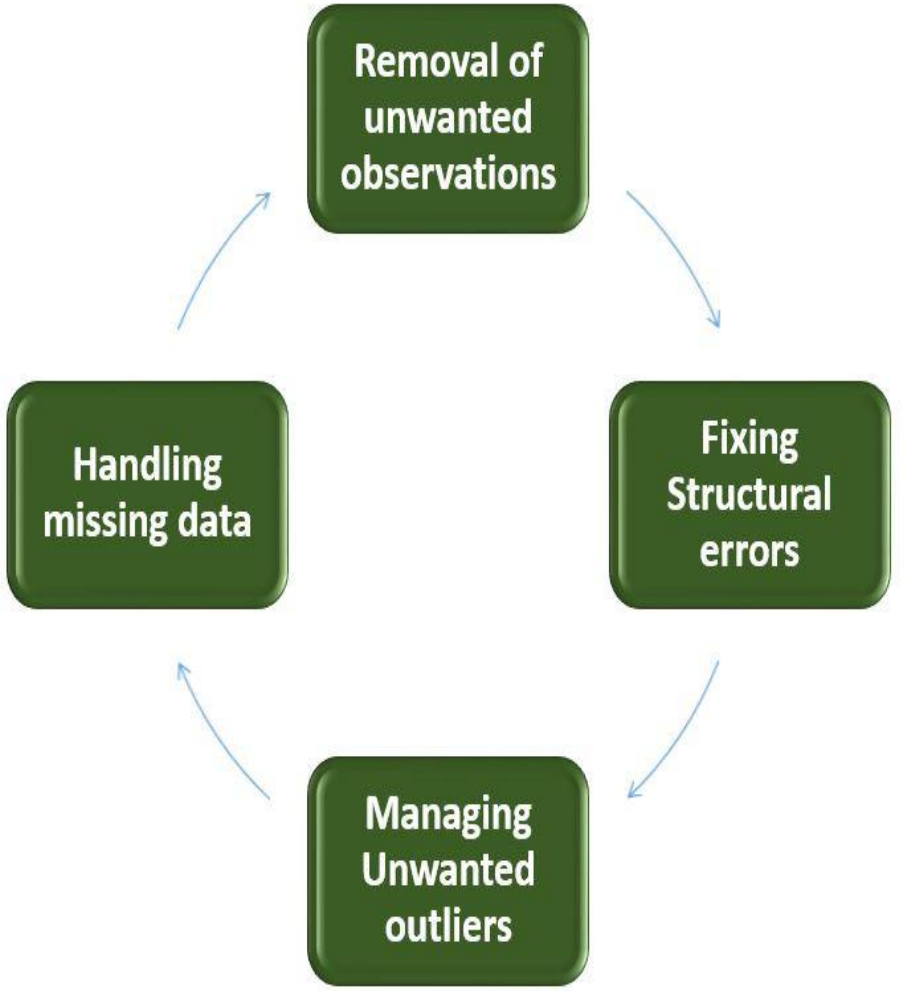
Data Cleaning in Data Science

- In the field of data science, where insights and predictions are drawn from vast and complex datasets, the quality of the input data significantly influences the validity of analytical results.
- Data cleaning involves the systematic identification and correction of errors, inconsistencies, and inaccuracies within a dataset, encompassing tasks such as handling missing values, removing duplicates, and addressing outliers.

Steps to Perform Data Cleanliness

Handling Missing Data:

- Devise strategies to handle missing data effectively.
- This may involve imputing missing values based on statistical methods, removing records with missing values, or employing advanced imputation techniques.
- Handling missing data ensures a more complete dataset, preventing biases and maintaining the integrity of analyses.



How to handle Missing Data

X1	X2	X3	X4	Y
78.5	67	1	0.2	73.2
76.5	67	0	0.2	69.2
78.5	67	0	0.2	69
78.5		0	0.2	69
75.5	66.5	1	0.2	73.5
75.5	66.5	1	0.4	
75.5	66.5	0	0.3	65.5
75.5	66.5	0	0.2	65.5
75		1		71
75	64	0	0.1	68
75	64	1	0.2	70.5

Ignore the Tuple

X1	X2	X3	X4	Y
78.5	67	1	0.2	73.2
76.5	67	0	0.2	69.2
78.5	67	0	0.2	69
~~78.5~~		~~0~~	~~0.2~~	~~69~~
75.5	66.5	1	0.2	73.5
~~75.5~~	~~66.5~~	~~1~~	~~0.4~~	
75.5	66.5	0	0.3	65.5
75.5	66.5	0	0.2	65.5
75		1		71
75	64	0	0.1	68
75	64	1	0.2	70.5

X1	X2	X3	X4	Y
78.5	67	1	0.2	73.2
76.5	67	0	0.2	69.2
78.5	67	0	0.2	69
75.5	66.5	1	0.2	73.5
75.5	66.5	0	0.3	65.5
75.5	66.5	0	0.2	65.5
75	64	0	0.1	68
75	64	1	0.2	70.5

How to handle Missing Data



X1	X2	X3	X4	Y
78.5	67	1	0.2	73.2
76.5	67	0	0.2	69.2
78.5	67	0	0.2	69
78.5	“unknown”	0	0.2	69
75.5	66.5	1	0.2	73.5
75.5	66.5	1	0.4	”unknown”
75.5	66.5	0	0.3	65.5
75.5	66.5	0	0.2	65.5
75	“unknown”	1	“unknown”	71
75	64	0	0.1	68
75	64	1	0.2	70.5

How to handle Missing Data



X1	X2	X3	X4	Y
78.5	67	1	0.2	73.2
76.5	67	0	0.2	69.2
78.5	67	0	0.2	69
78.5	66.1	0	0.2	69
75.5	66.5	1	0.2	73.5
75.5	66.5	1	0.4	“69.44”
75.5	66.5	0	0.3	65.5
75.5	66.5	0	0.2	65.5
75	“66.1”	1	“0.22”	71
75	64	0	0.1	68
75	64	1	0.2	70.5

How to handle Missing Data

The k-nearest neighbors (KNN) algorithm is a non-parametric, supervised learning classifier, which uses proximity to make classifications or predictions about the grouping of an individual data point.

X1	X2	X3	X4	Y
78.5	67	1	0.2	73.2
76.5	67	0	0.2	69.2
78.5	67	0	0.2	69
78.5	66.1	0	0.2	69
75.5	66.5	1	0.2	73.5
75.5	66.5	1	0.4	“69.44”
75.5	66.5	0	0.3	65.5
75.5	66.5	0	0.2	65.5
75	“66.1”	1	“0.22”	71
75	64	0	0.1	68
75	64	1	0.2	70.5

Data Cleanliness - Noisy Data



- Binning
  - Binning methods smooth a sorted data value by consulting its "neighbor-hood," that is, the values around it.
  - first sort data and partition into (equal-frequency) bins
  - then one can *smooth by bin means, smooth by bin median, smooth by bin boundaries,* etc.
- Regression
  - smooth by fitting the data into regression functions
- Clustering
  - detect and remove outliers
- Semi-automated method: combined computer and human inspection
  - detect suspicious values and check manually

Binning Methods for Data Smoothing



- **Sorted data for price (in dollars):**

  4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

* **Partition into equal-frequency (equi-depth) bins:**

  - Bin 1: 4, 8, 9, 15

  - Bin 2: 21, 21, 24, 25

  - Bin 3: 26, 28, 29, 34

- **Smoothing by bin means:** each value in a bin is replaced by the mean value of the bin.

  - Bin 1: 9, 9, 9, 9

  - Bin 2: 23, 23, 23, 23

  - Bin 3: 29, 29, 29, 29

Binning Methods for Data Discretization using Python




```
import pandas as pd

# Sample DataFrame
data = {'Rating': [2.5, 3.8, 4.5, 1.2, 4.0, 2.2, 3.0, 4.8, 1.9, 2.9]}
df = pd.DataFrame(data)

# Define the bins and labels
bins = [0, 2, 3.5, 5]  # Bin edges (0-2: Low, 2-3.5: Medium, 3.5-5: High)
labels = ['Low', 'Medium', 'High']

# Discretize the Rating column
df['Rating_Category'] = pd.cut(df['Rating'], bins=bins, labels=labels,
include_lowest=True)

# Display the DataFrame
print(df)
```

Binning Methods for Data Smoothing



Equal-width partitioning

$$\text{width} = \frac{\max - \min}{\text{number of bins}} = \frac{215 - 5}{3} = 70.$$

$$5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215$$

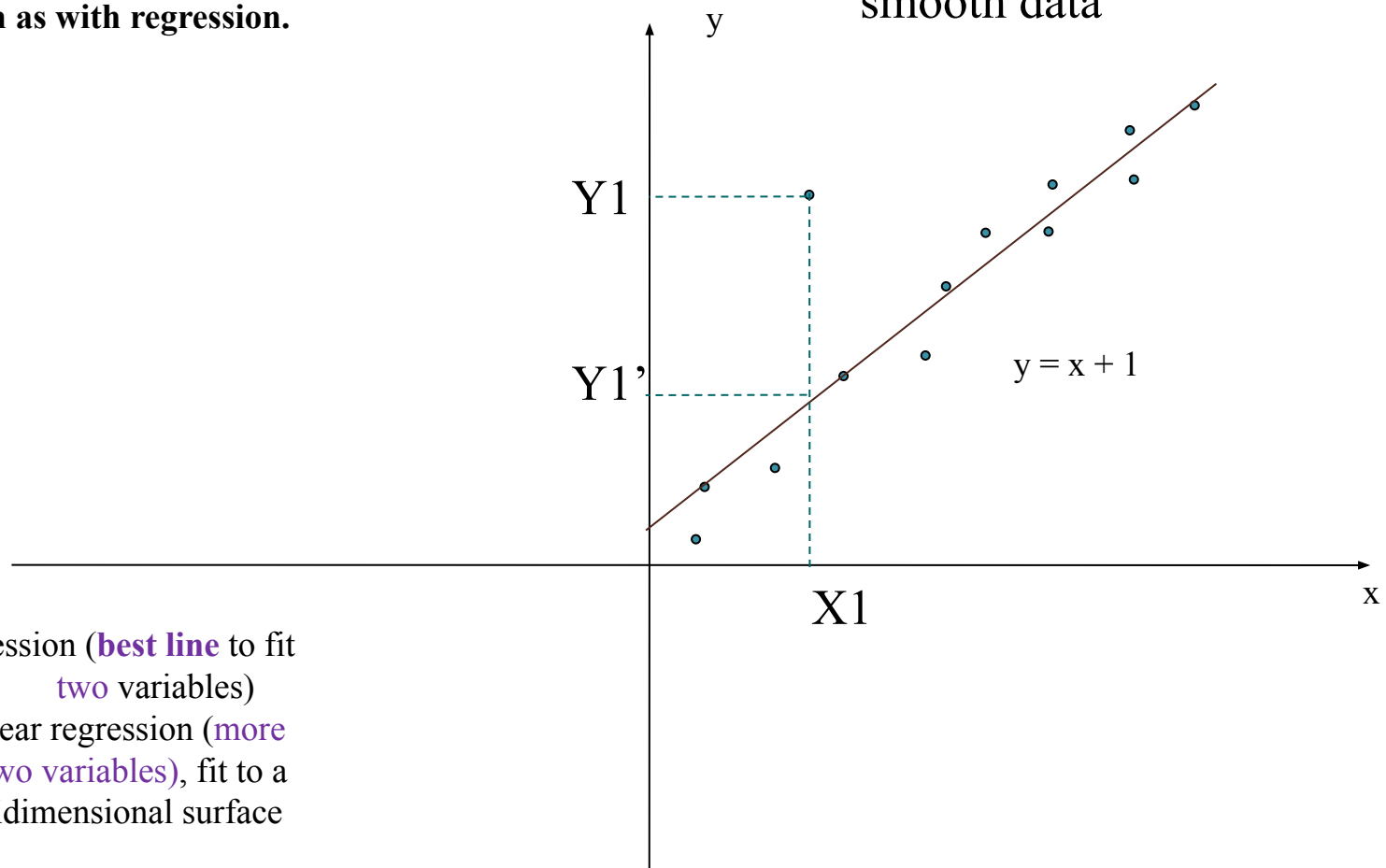
1. $70 + 5 = 75$ (from 5 to 75) =Bin $1 : 5, 10, 11, 13, 15, 35, 50, 55, 72$
2. $75 + 70 = 145$ (from 75 to 145) =Bin $2 : 92$
3. $145 + 70 = 215$ (from 145 to 215) =Bin $3 : 204, 215$

Binning Methods for Data Smoothing

Regression

Linear regression – find the best line to fit two variables and use regression function to smooth data

Data can be smoothed by fitting the data to a function, such as with regression.

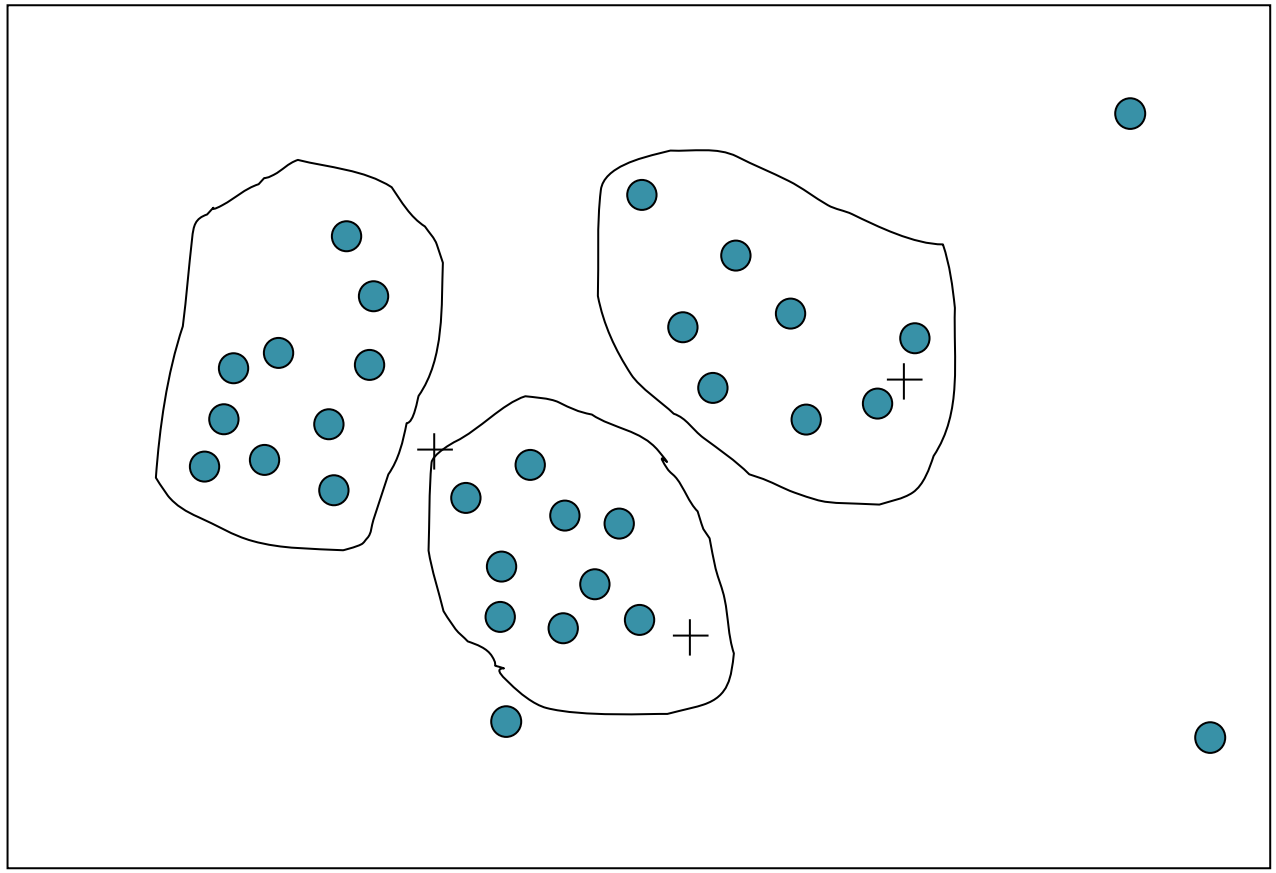


- Linear regression (**best line** to fit two variables)
- Multiple linear regression (more than two variables), fit to a multidimensional surface

Binning Methods for Data Smoothing

Cluster Analysis

detect and remove outliers, Where similar values are organized into groups or “clusters”



Binning Methods for Data Smoothing

- **Sorted data for price (in dollars):**

  4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

* **Partition into equal-frequency (equi-depth) bins:**
  - **Bin 1: 4, 8, 9, 15**
  - **Bin 2: 21, 21, 24, 25**
  - **Bin 3: 26, 28, 29, 34**

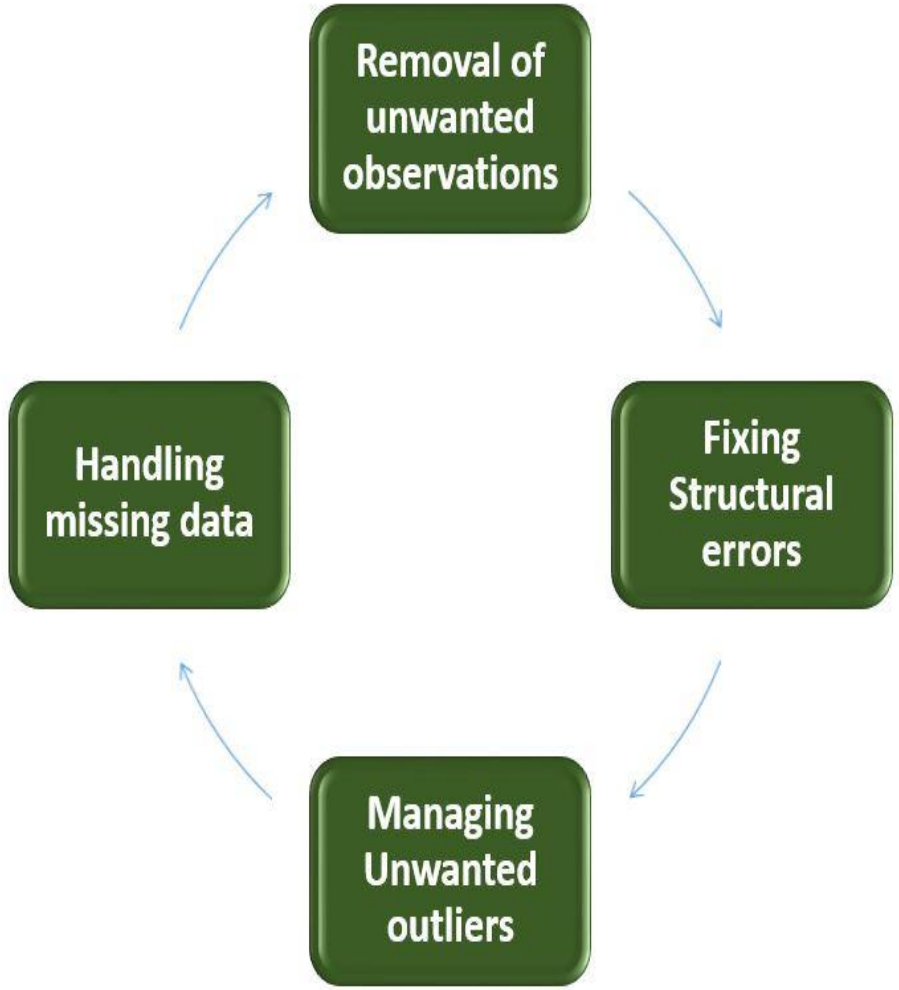
- **Smoothing by bin means:** each value in a bin is replaced by the mean value of the bin.

  - **Bin 1: 9, 9, 9, 9**
  - **Bin 2: 23, 23, 23, 23**
  - **Bin 3: 29, 29, 29, 29**

Steps to Perform Data Cleanliness

Removal of Unwanted Observations:

- Identify and eliminate irrelevant or redundant observations from the dataset.
- The step involves
  - scrutinizing data entries for duplicate records
  - irrelevant information
  - data points that do not contribute meaningfully to the analysis.
- Removing unwanted observations streamlines the dataset, reducing noise and improving the overall quality.



Removal of Unwanted Observations



Removal of Unwanted Observations:

This includes deleting duplicate/ redundant or irrelevant values from your dataset.

- Redundant observations alter the efficiency to a great extent as the data repeats and may add towards the correct side or towards the incorrect side, thereby producing unfaithful results.
- Irrelevant observations are any type of data that is of no use to us and can be removed directly.

PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
0 1 0 3 Braund, Mr. Owen Harris male 22.0 1 0 A/5 21171 7.2500 NaN S
1 2 1 1 Cumings, Mrs. John Bradley (Florence Briggs Th... female 38.0 1 0 PC 17599 71.2833 C85 C
2 3 1 3 Heikkinen, Miss. Laina female 26.0 0 0 STON/O2. 3101282 7.9250 NaN S
3 4 1 1 Futrelle, Mrs. Jacques Heath (Lily May Peel) female 35.0 1 0 113803 53.1000 C123 S
4 5 0 3 Allen, Mr. William Henry male 35.0 0 0 373450 8.0500 NaN S

```
# Categorical columns
cat_col = [col for col in df.columns if df[col].dtype == 'object']
print('Categorical columns :',cat_col)
# Numerical columns
num_col = [col for col in df.columns if df[col].dtype != 'object']
print('Numerical columns :',num_col)
```

```
Categorical columns : ['Name', 'Sex', 'Ticket', 'Cabin', 'Embarked']
Numerical columns : ['PassengerId', 'Survived', 'Pclass', 'Age', 'SibSp',
'Parch', 'Fare']
```



Removal of Unwanted Observations



Removal of Unwanted Observations:

This includes deleting duplicate/ redundant or irrelevant values from your dataset.

- Redundant observations alter the efficiency to a great extent as the data repeats and may add towards the correct side or towards the incorrect side, thereby producing unfaithful results.
- Irrelevant observations are any type of data that is of no use to us and can be removed directly.

```
# Categorical columns
cat_col = [col for col in df.columns if df[col].dtype == 'object']
print('Categorical columns :',cat_col)
# Numerical columns
num_col = [col for col in df.columns if df[col].dtype != 'object']
print('Numerical columns :',num_col)
```

```
Categorical columns : ['Name', 'Sex', 'Ticket', 'Cabin', 'Embarked']
Numerical columns : ['PassengerId', 'Survived', 'Pclass', 'Age', 'SibSp',
'Parch', 'Fare']
```

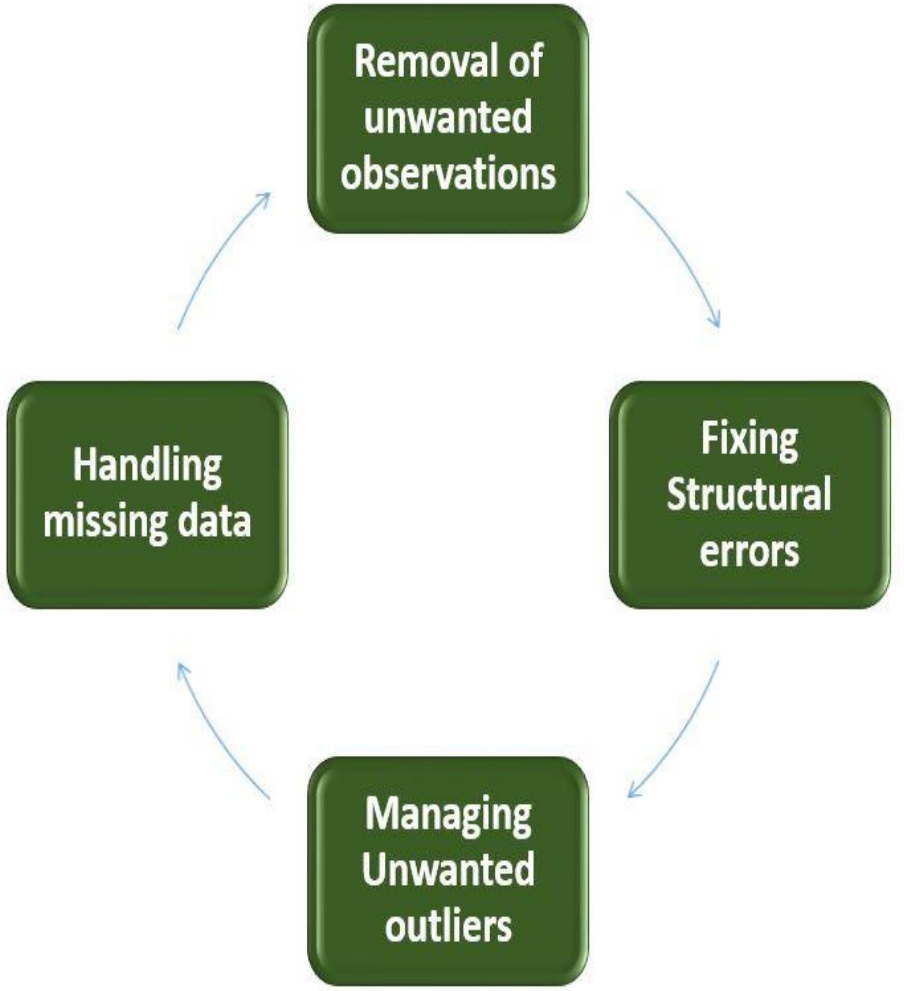
As we know our machines don't understand the text data. So, we have to either drop or convert the categorical column values into numerical types. Here we are dropping the Name columns because the Name will be always unique and it hasn't a great influence on target variables.

Microsoft Excel
97-2004

Steps to Perform Data Cleanliness

Fixing Structure errors:

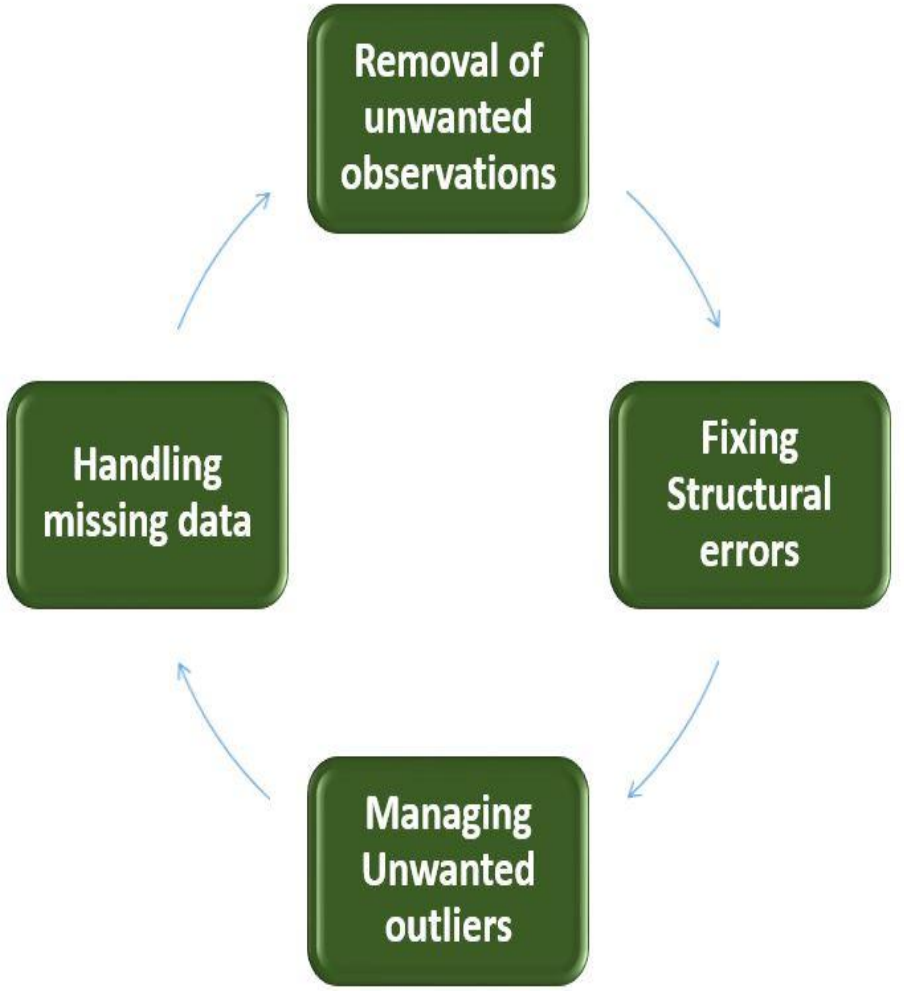
- Address structural issues in the dataset
  - inconsistencies in data formats
  - naming conventions
  - variable types
- Standardize formats, correct naming discrepancies, an ensure uniformity in data representation.
- Fixing structure errors enhances data consistency an facilitates accurate analysis and interpretation.



Steps to Perform Data Cleanliness

Managing Unwanted Outliers:

- Identify and manage outliers, which are data points significantly deviating from the norm.
- Depending on the context, decide whether to remove outliers or transform them to minimize their impact on analysis.
- Managing outliers is crucial for obtaining more accurate and reliable insights from the data.
- Managing outliers ensures that extreme values do not unduly influence analysis.



Managing Unwanted Outliers

Handling Outliers

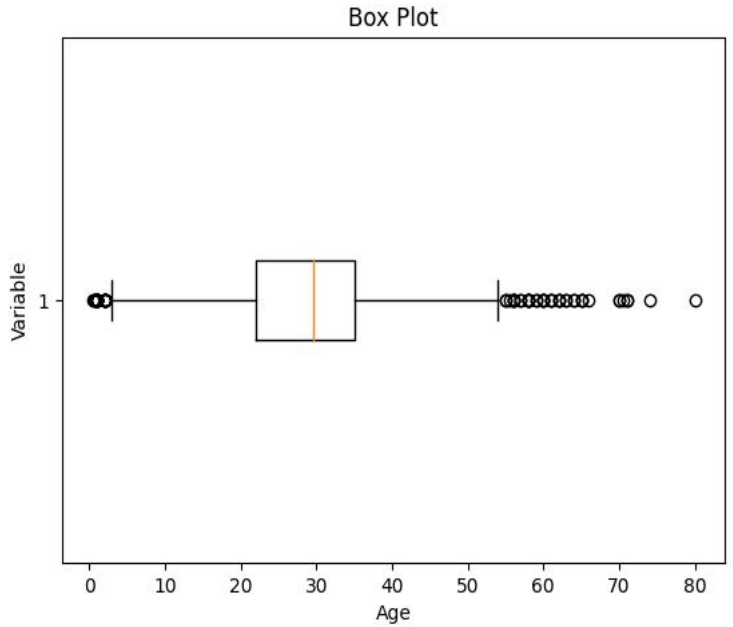
- Outliers are extreme values that deviate significantly from the majority of the data.
- They can negatively impact the analysis and model performance. Techniques such as clustering, interpolation, or transformation can be used to handle outliers.

To check the outliers,

- We generally use a box plot. A box plot, also referred to as a box-and-whisker plot, is a graphical representation of a dataset's distribution.
- It shows a variable's median, quartiles, and potential outliers. The line inside the box denotes the median, while the box itself denotes the interquartile range (IQR).
- The whiskers extend to the most extreme non-outlier values within 1.5 times the IQR. Individual points beyond the whiskers are considered potential outliers.
- A box plot offers an easy-to-understand overview of the range of the data and makes it possible to identify outliers or skewness in the distribution.

Let's plot the box plot for Age column data.

```
import matplotlib.pyplot as plt
plt.boxplot(df3['Age'], vert=False)
plt.ylabel('Variable')
plt.xlabel('Age')
plt.title('Box Plot')
plt.show()
```



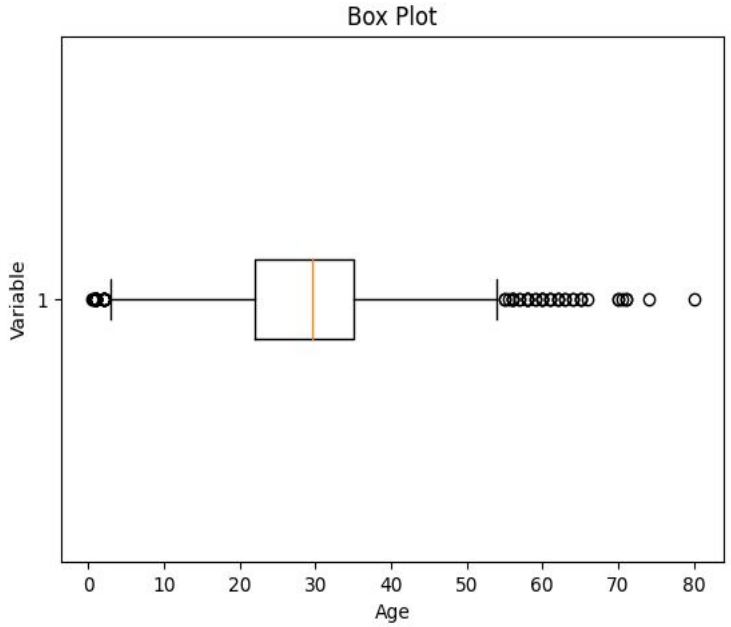
As we can see from the above Box and whisker plot, Our age dataset has outliers values. The values less than 5 and more than 55 are outliers

Microsoft Excel 97-2004

Managing Unwanted Outliers

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```



As we can see from the above Box and whisker plot, Our age dataset has outliers values. The values less than 5 and more than 55 are outliers

```
# calculate summary statistics
mean = df3['Age'].mean()
std  = df3['Age'].std()

# Calculate the lower and upper bounds
lower_bound = mean - std*2
upper_bound = mean + std*2

print('Lower Bound :',lower_bound)
print('Upper Bound :',upper_bound)

# Drop the outliers
df4 = df3[(df3['Age'] >= lower_bound)
          & (df3['Age'] <= upper_bound)]
```

Lower Bound : 3.705400107925648
Upper Bound : 55.578785285332785

Similarly, we can remove the outliers of the remaining columns.

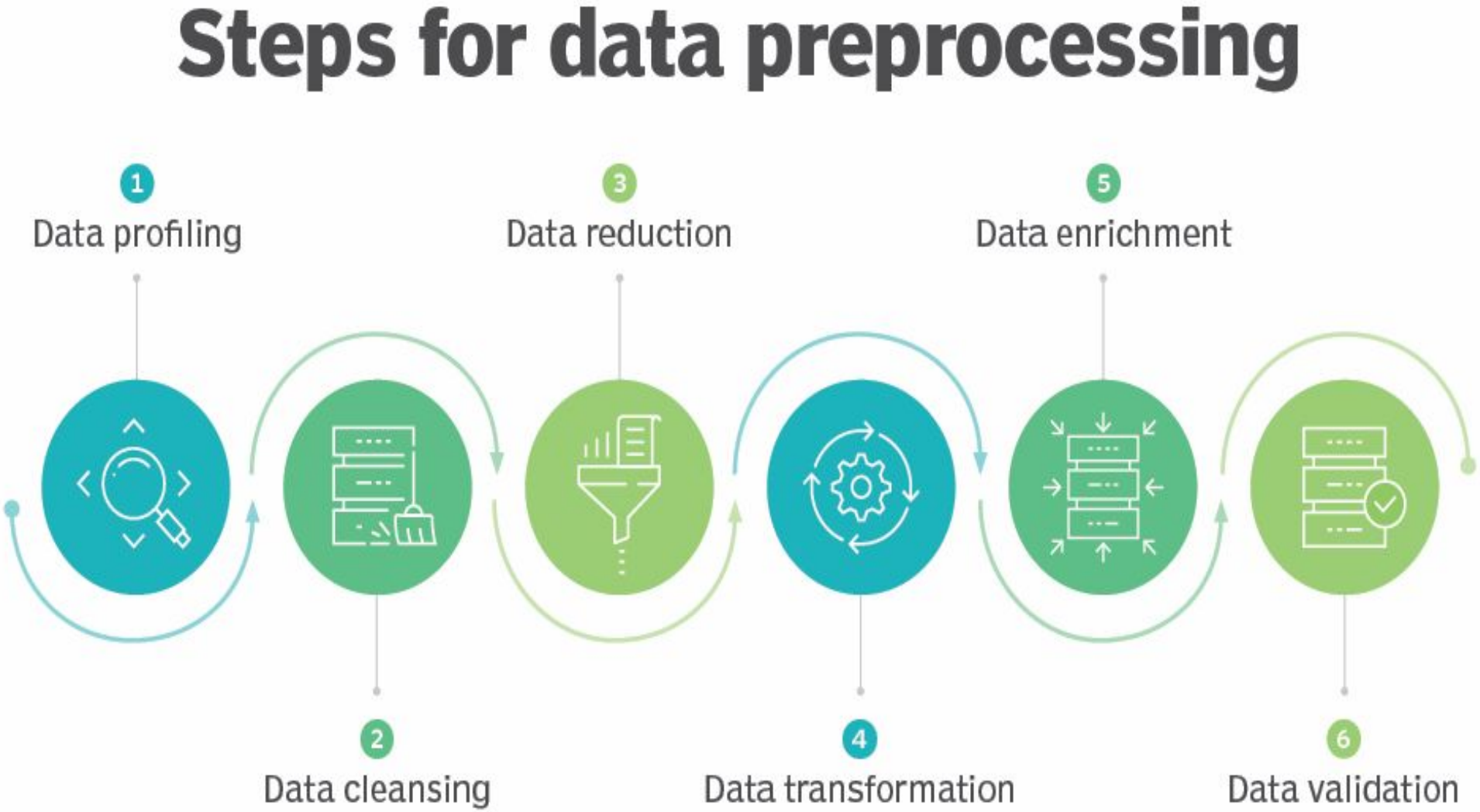
Microsoft Excel 97-2004

Data Cleaning– some videos



- https://www.youtube.com/watch?v=nijQ8ZiHv58
- https://www.youtube.com/watch?v=kNl7YDN-_js
- https://www.youtube.com/watch?v=4wxQC1pbQig

Key steps in Data Pre-processing



Data Reduction

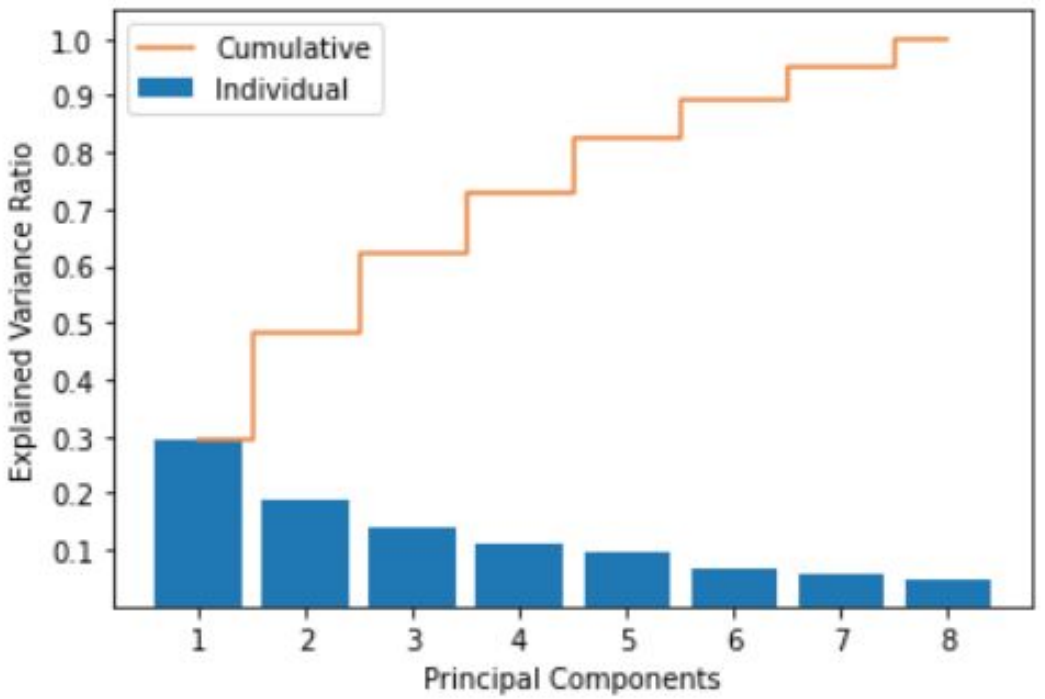
- Data Reduction is used to reduce the volume or size of the input data.
- To reduce storage and analysis costs and improve storage efficiency

Dimensionality Reduction

- Process of reducing the number of features in the input dataset.
- Selecting features with the highest importance, Principal Component Analysis (PCA), etc.

Data Compression

- Data is compressed.
- Data Compression can be lossless or lossy depending on whether the information is lost or not during compression.



The higher the number of Principal Components retained, the higher is the variance preserved.

Reduced our data set dimensionality from 8 columns to 6 columns

```
pca = PCA(n_components=6)

X_train_pca = pca.fit_transform(X_train_normal)
X_test_pca = pca.fit_transform(X_test_normal)

print("Train Set columns: ", X_train_pca.shape[1])
print("Test Set Columns: ", X_test_pca.shape[1])
```

Data Reduction



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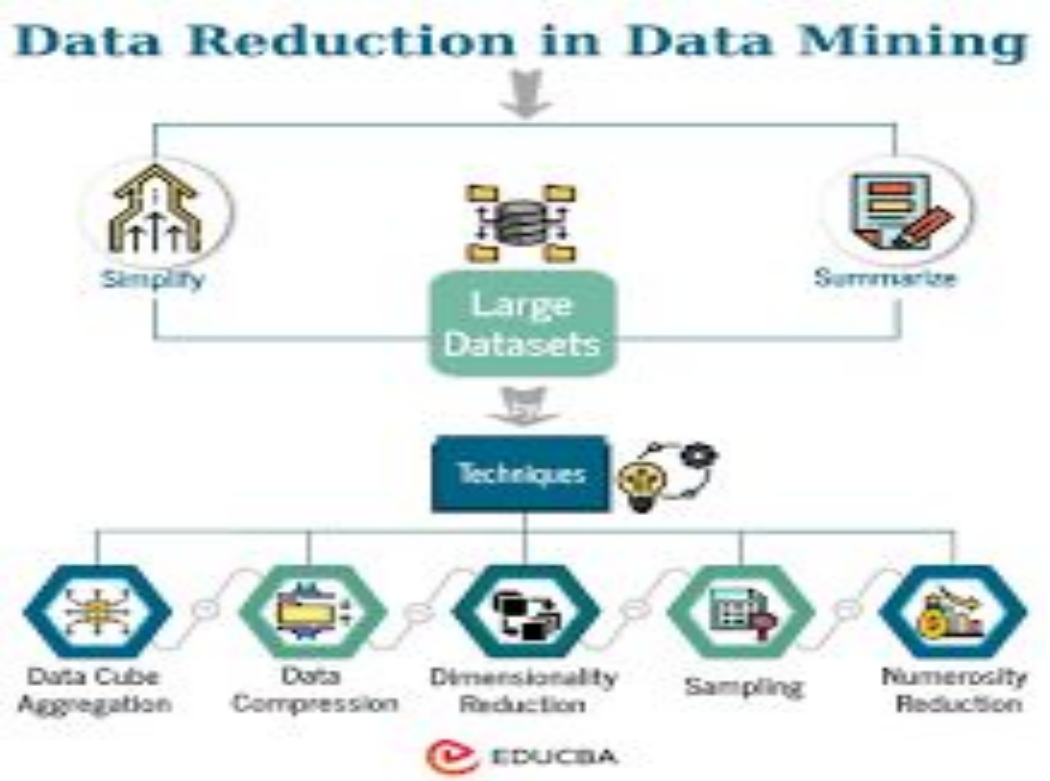
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print("Train Set columns: ", X_train_pca.shape[1])
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```

Data Reduction

- Data reduction is a technique used in data mining to reduce the size of a dataset while still preserving the most important information.
- This can be beneficial in situations where the dataset is too large to be processed efficiently, or where the dataset contains a large amount of irrelevant or redundant information.



Data Reduction – Dimension Reduction



Dimension reduction:
Whenever we come across any data which is weakly important, then we use the attribute required for our analysis. It reduces data size as it eliminates outdated or redundant features.

Step-wise Forward Selection –
The selection begins with an empty set of attributes later on we decide the best of the original attributes on the set based on their relevance to other attributes. We know it as a p-value in statistics. Suppose there are the following attributes in the data set in which few attributes are redundant.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}
Initial reduced attribute set: { }
Step-1: {X1}
Step-2: {X1, X2}
Step-3: {X1, X2, X5}
Final reduced attribute set: {X1, X2, X5}

Data Reduction – Dimension Reduction



Dimension reduction:
Whenever we come across any data which is weakly important, then we use the attribute required for our analysis. It reduces data size as it eliminates outdated or redundant features.

Step-wise Backward Selection –
This selection starts with a set of complete attributes in the original data and at each point, it eliminates the worst remaining attribute in the set.

Suppose there are the following attributes in the data set in which few attributes are redundant.

Initial attribute Set: {X1, X2, X3, X4, X5, X6}
Initial reduced attribute set: {X1, X2, X3, X4, X5, X6 }
Step-1: {X1, X2, X3, X4, X5}
Step-2: {X1, X2, X3, X5}
Step-3: {X1, X2, X5}
Final reduced attribute set: {X1, X2, X5}

Data Reduction – Data Compression

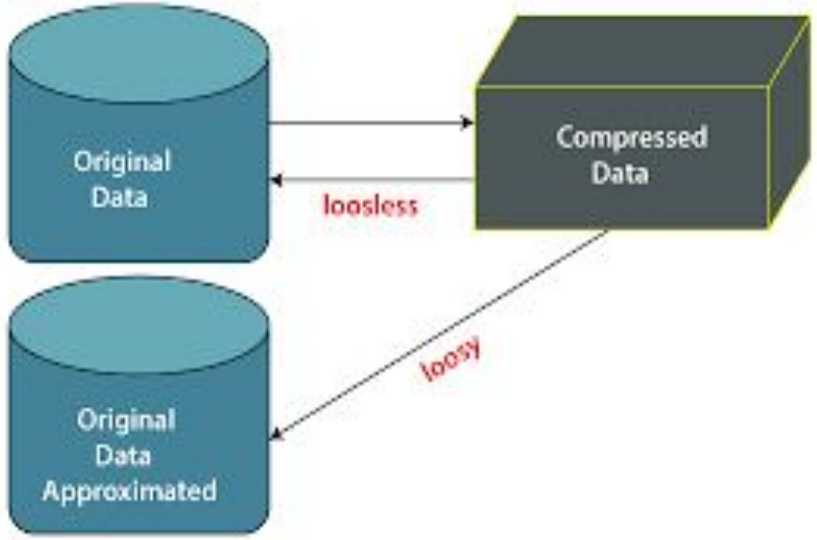
The data compression technique reduces the size of the files using different encoding mechanisms (Huffman Encoding & run-length Encoding). We can divide it into two types based on their compression techniques.

•Lossless Compression –

Encoding techniques (Run Length Encoding) allow a simple and minimal data size reduction. Lossless data compression uses algorithms to restore the precise original data from the compressed data.

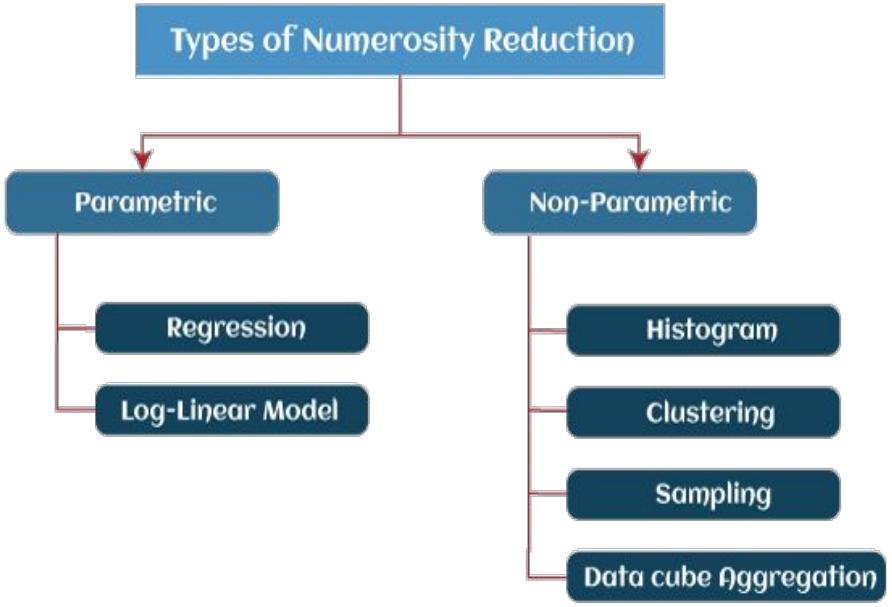
•Lossy Compression –

Methods such as the Discrete Wavelet transform technique, PCA (principal component analysis) are examples of this compression. In lossy-data compression, the decompressed data may differ from the original data but are useful enough to retrieve information from them.



Data Reduction – Numerosity

In this reduction technique, the actual data is replaced with mathematical models or smaller representations of the data instead of actual data, it is important to only store the model parameter.
Or non-parametric methods such as clustering, histogram, and sampling.



For numeric data following techniques can be followed:
Histogram analysis –
The histogram is used to partition the value for the attribute X, into disjoint ranges called brackets. There are several partitioning rules:

- Equal Frequency partitioning: Partitioning the values based on their number of occurrences in the data set.
- Equal Width Partitioning: Partitioning the values in a fixed gap based on the number of bins i.e. a set of values ranging from 0-20.
- Clustering: Grouping similar together

Data Reduction – Advantages / Disadvantages

Advantages

- Improved efficiency: Data reduction can help to improve the efficiency of machine learning algorithms by reducing the size of the dataset.
- Improved performance: Data reduction can help to improve the performance of machine learning algorithms by removing irrelevant or redundant information from the dataset.
- Reduced storage costs: Data reduction can help to reduce the storage costs associated with large datasets by reducing the size of the data.
- Improved interpretability: Data reduction can help to improve the interpretability of the results by removing irrelevant or redundant information from the dataset.

Disadvantages

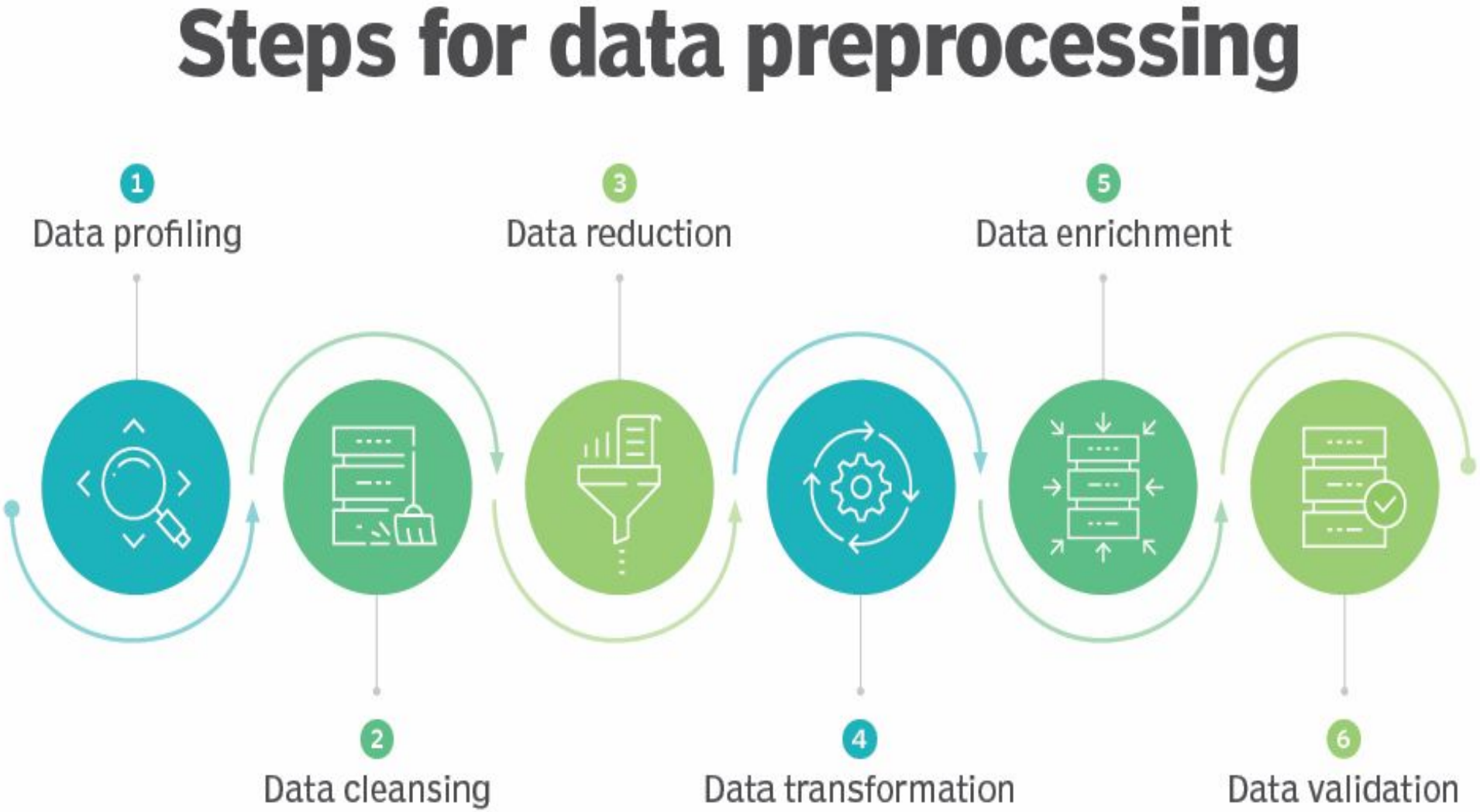
- Loss of information: Data reduction can result in a loss of information, if important data is removed during the reduction process.
- Impact on accuracy: Data reduction can impact the accuracy of a model, as reducing the size of the dataset can also remove important information that is needed for accurate predictions.
- Impact on interpretability: Data reduction can make it harder to interpret the results, as removing irrelevant or redundant information can also remove context that is needed to understand the results.
- Additional computational costs: Data reduction can add additional computational costs to the data mining process, as it requires additional processing time to reduce the data.

Data Reduction– some videos



- https://www.youtube.com/watch?v=NAL_3fuPfQQ
- https://www.youtube.com/watch?v=M65LHFdt63I
- https://www.youtube.com/watch?v=Mtherb-jBR8

Key steps in Data Pre-processing



Data Transformation

- Data transformation techniques refer to all the actions that help you transform your **raw data into a clean and ready-to-use dataset**.
- The goal of data transformation is to prepare the data for data mining so that it can be used to extract useful insights and knowledge.
- It is also a critical component for any organization seeking to leverage its data to generate timely business insights.
- Data transformation -- when done properly -- it ensures data is easy to access, consistent, secure and ultimately trusted by the intended business users.



Data Transformation - Techniques

A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values

Methods

- Smoothing: Remove noise from data
- Attribute/feature construction

  New attributes constructed from the given ones
- Aggregation: Summarization, data cube construction
- Normalization: Scaled to fall within a smaller, specified range

  min-max normalization

  z-score normalization

  normalization by decimal scaling
- Discretization: Concept hierarchy climbing
- Generalization. It converts low-level data attributes to high-level data attributes using concept hierarchy.



Data Transformation – Soothing (Already discussed in Cleaning)

- It is a process that is used to remove noise from the dataset using some algorithms
- It allows for highlighting important features present in the dataset.
- It helps in predicting the patterns.

Methods

- Binning
- Histogram Analysis
- Clustering



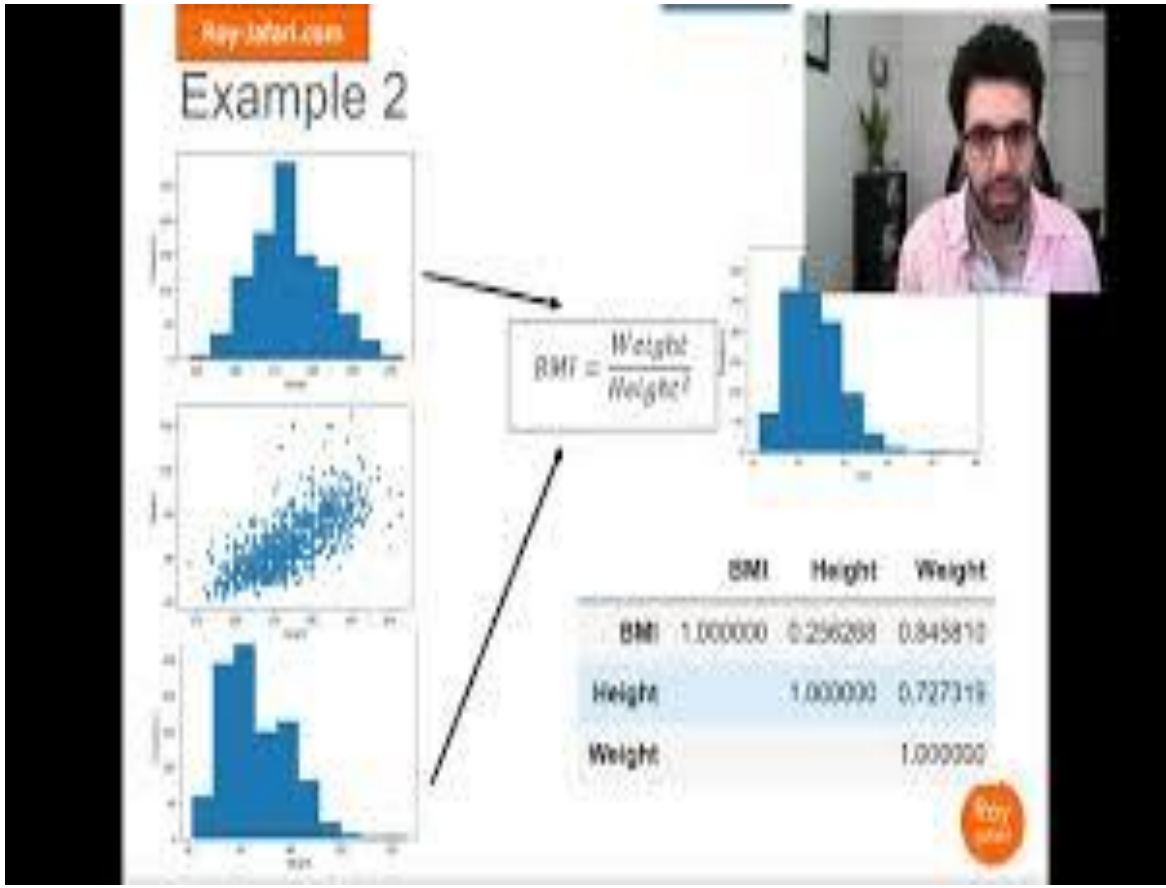
Data Transformation – Attribute Construction

Where new attributes are created & applied to assist the mining process from the given set of attributes. This simplifies the original data & makes the mining more efficient.

For example, suppose we have a data set referring to measurements of different plots, i.e., we may have the height and width of each plot.

So here, we can construct a new attribute 'area' from attributes 'height' and 'weight'.

This also helps understand the relations among the attributes in a data set.



Data Transformation - Generalization

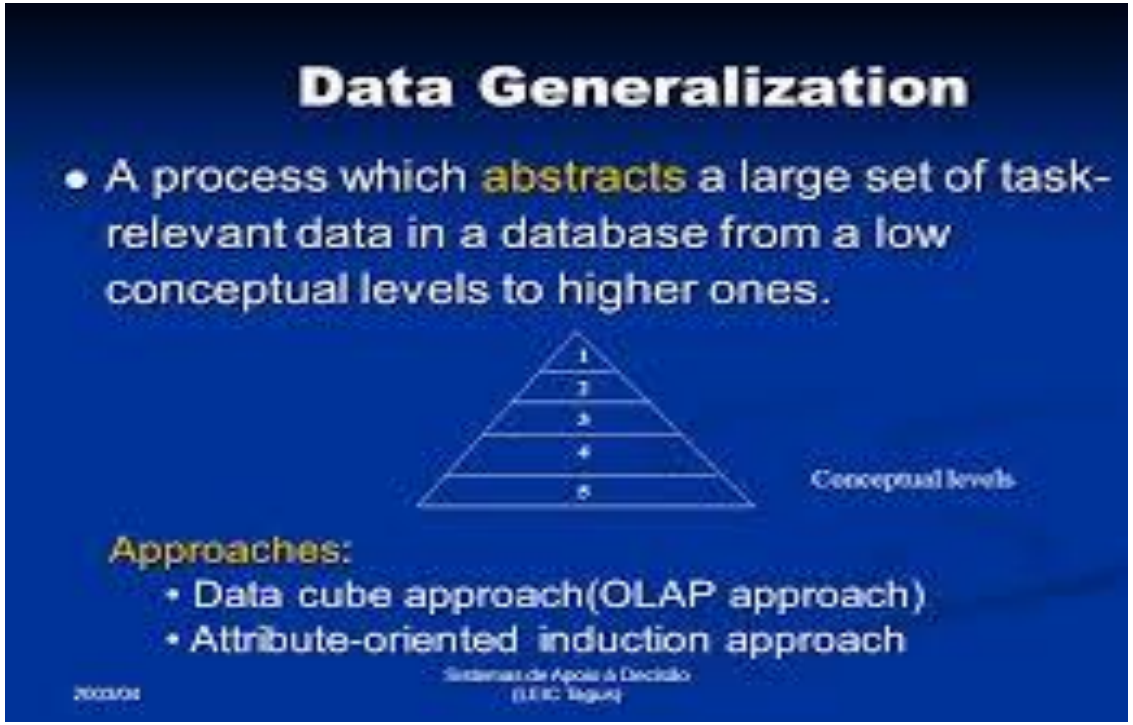
It converts low-level data attributes to high-level data attributes using concept hierarchy.

For Example Age initially in Numerical form (22, 25) is converted into categorical value (young, old).

For **example**, Categorical attributes, such as house addresses, may be generalized to higher-level definitions, such as town or country.

Data generalization can be divided into two approaches:

- Data cube process (OLAP) approach.
- Attribute-oriented induction (AOI) approach.



Data Transformation - Generalization

Data cube approach :

Is also known as OLAP approach.

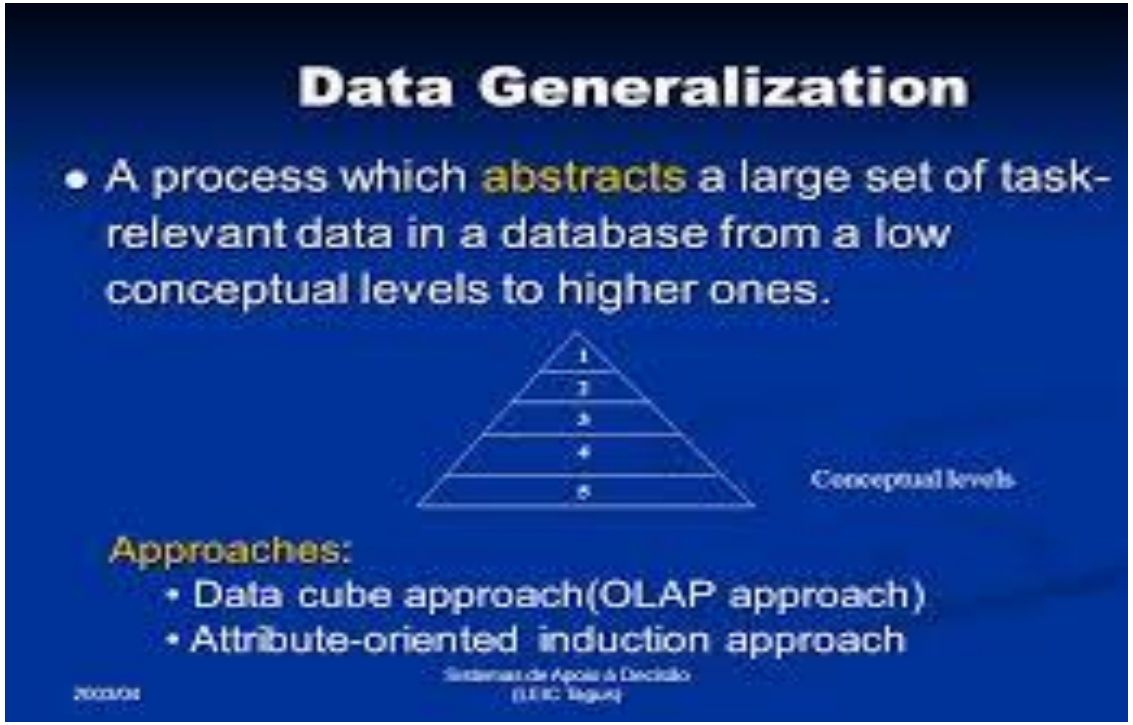
•It is an efficient approach as it is helpful to make the past selling graph.

•In this approach, computation and results are stored in the Data cube.

•It uses Roll-up and Drill-down operations on a data cube.

•These operations typically involve aggregate functions, such as count(), sum(), average(), and max().

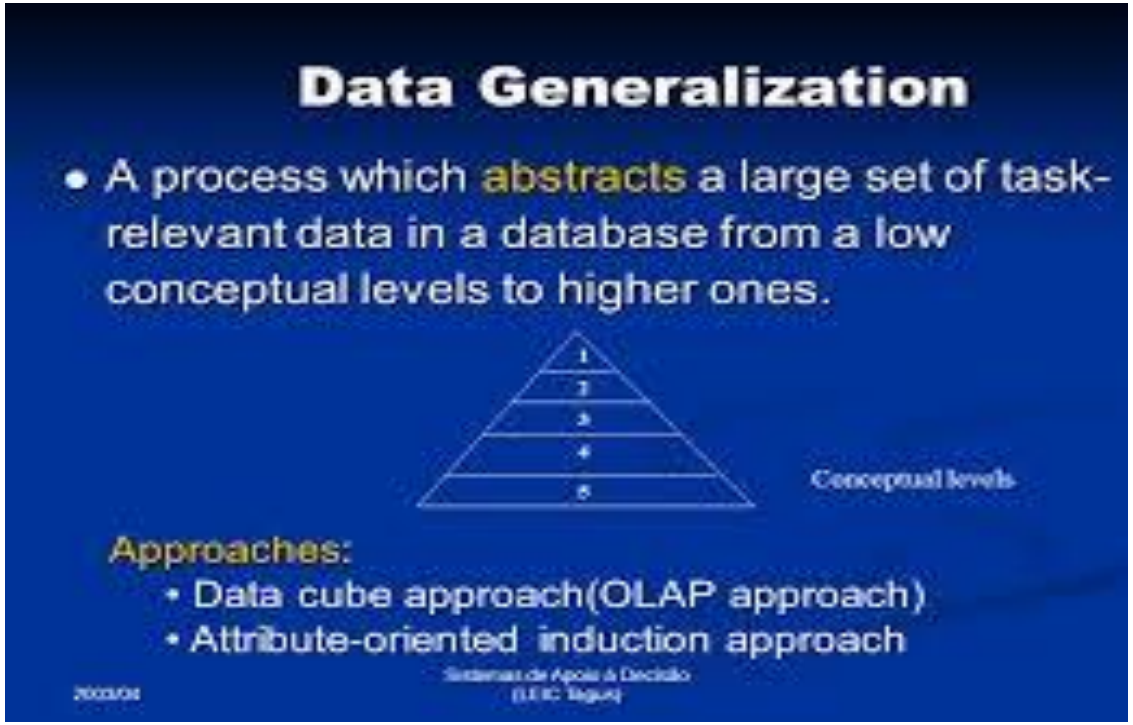
•These materialized views can then be used for decision support, knowledge discovery, and many other applications.



Data Transformation - Generalization

Attribute Oriented Induction Approach

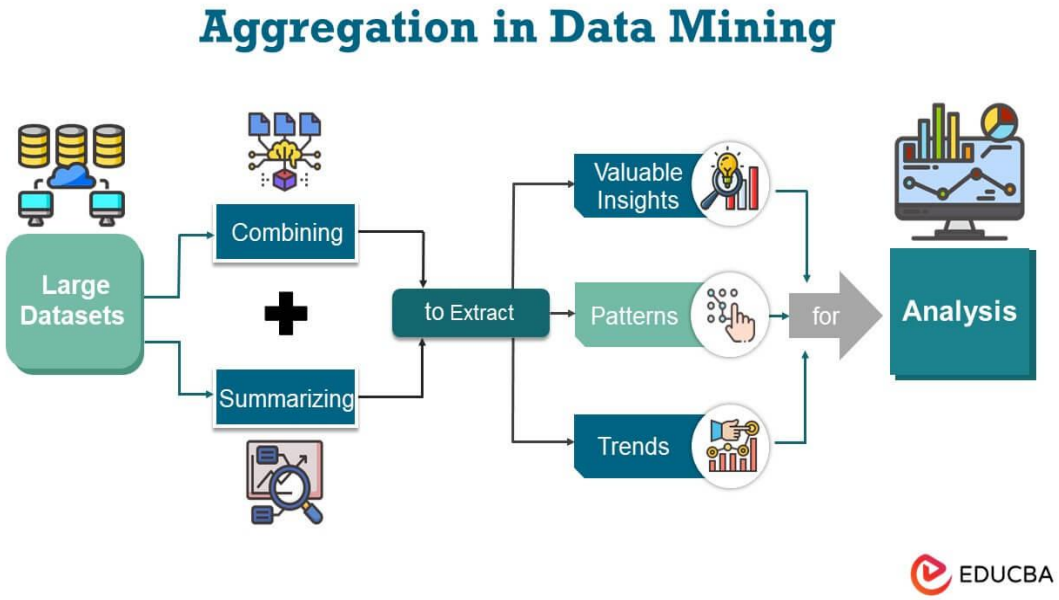
- It is an online data analysis, query oriented and generalization based approach.
- In this approach, we perform generalization on basis of different values of each attributes within the relevant data set. after that same tuple are merged and their respective counts are accumulated in order to perform aggregation.
- It performs off-line aggregation before an OLAP or data mining query is submitted for processing.



Data Transformation - Aggregation

Data Aggregation

- Aggregation is the process of summarizing and condensing large datasets for analysis.
- It involves grouping data based on specific attributes or criteria.
- Mathematical functions like sum, average, or count are commonly used for aggregation.
- Aggregation simplifies complex data, making it easier to identify patterns and make informed decisions.
- It provides a comprehensive overview of data, revealing essential insights for data analysis and decision-making.



Data Transformation - Aggregation

Types of Data Aggregation

- **Time Aggregation:** Aggregating data over specific time intervals, such as daily, weekly, or monthly, to analyze trends and patterns over time.
- **Spatial Aggregation:** Combining data based on geographical regions or spatial units, facilitating geographic analysis and visualization.

Techniques for Aggregation

- Summation: This function adds up values within a group, providing the total.
- Averaging: Averages the values within a group, yielding the mean.
- Counting: Counts the number of occurrences within a group, often used for tallying events like website visits or product sales.
- Max and Min: These functions find the highest (max) and lowest (min) values within a group, helping identify extremes, such as peak or minimum temperatures.

1. **Grouping and Group-by Operations:** Smaller subsets of data are created through specific criteria, such as product categories. They are common in SQL queries to aggregate data within these groups.
2. **Roll-up and Drill-down:** In hierarchical data structures, roll-up involves aggregating data from lower to higher levels, while drill-down does the opposite—breaking aggregated data into more detailed levels.
3. **Cube Aggregation:** This method is used in multidimensional data analysis. It involves summing up data across multiple dimensions or “slices” of a data cube to provide different perspectives on the data.
4. **Hierarchical Aggregation:** In hierarchical data, this method summarizes data at different levels of a hierarchy, allowing analysis at various levels of granularity, from broad to detailed views.

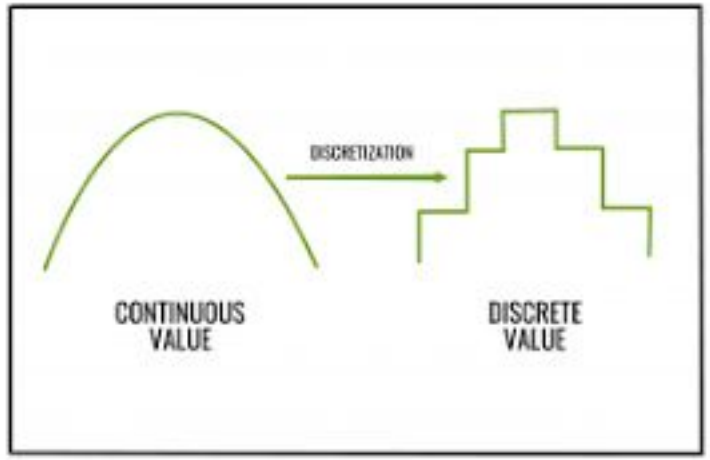
Data Transformation - Discretization

Data Discretization

- Transforming continuous data into a set of data intervals.

Ex:

- Imagine having tens of thousands of rows representing people in a survey providing their first name, last name, age, and gender.
- Age is a numerical attribute that can have a lot of different values. To make our life easier we can divide the range of this continuous attribute into intervals.
- youth, middle-aged, and senior



Data Transformation - Discretization

Discretization By Histogram:

Histogram analysis is unsupervised learning because it doesn't use any class information like binning.

There are various partition rules used to define histograms. In equal width histogram, values are partitioned in equal size bins or ranges. in our earlier example, we have created bin of size 25, which is an equal-width histogram.

In equal frequency histogram, partition is done in such a way that every bucket contains same number of data tuples

Example: The following data shows the price of commonly sold items in sorted order: 1,1,4,4,4,4,7,7,9,9,9,9,9,11, 13,13,13,17,17,17,17,17,17, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30,30, Following figure shows for current data

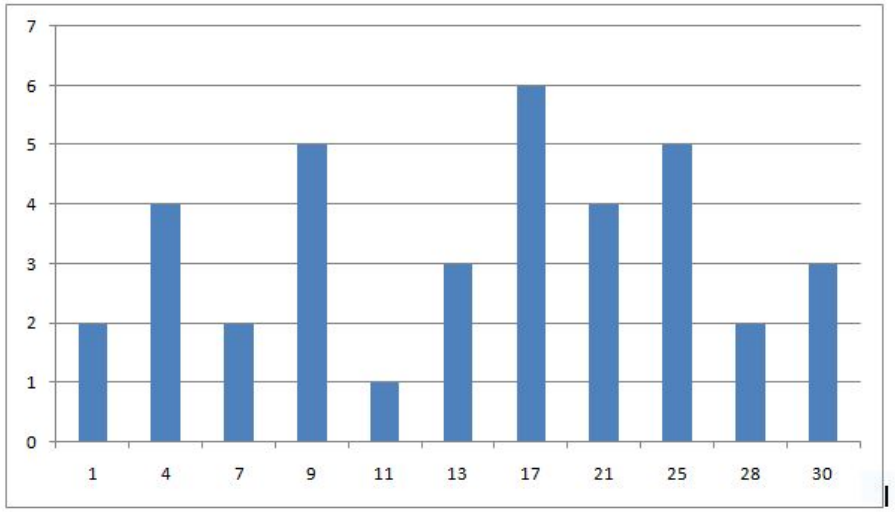


Figure 1 Histogram using price where one bucket represents one value

Now, we will partition into equal width bins where every bucket has same size width of 10.

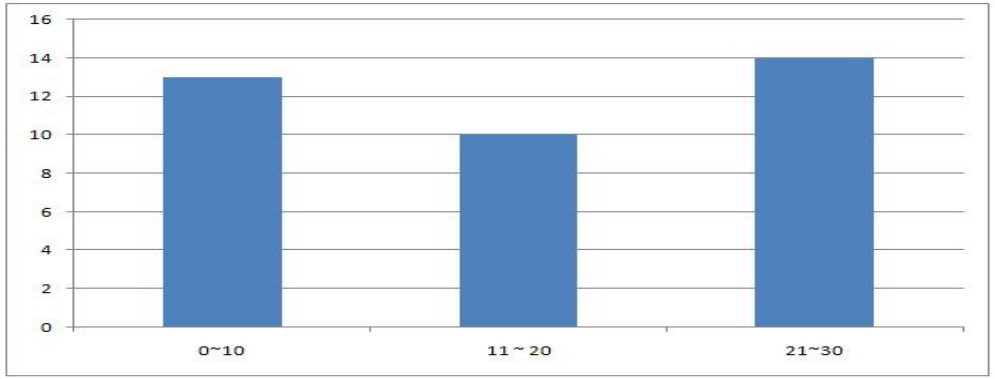


Figure 2 : Equal width Histogram

Data Transformation

Data Aggregation

- Storing and presenting data in a summary format.
- To perform statistical analysis of data as you might want to aggregate your data over a specific time period and provide statistics such as average, sum, minimum, and maximum.

Data Discretization

- Transforming continuous data into a set of data intervals.

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	A	B
1	**Date**	**Temperature, C**
2	1981-01-01	20.7
3	1981-01-02	17.9
4	1981-01-03	18.8
5	1981-01-04	14.6
6	1981-01-05	15.8
7	1981-01-06	15.8
8	1981-01-07	15.8
9	1981-01-08	17.4
10	1981-01-09	21.8
11	1981-01-10	20
12	1981-01-11	16.2
13	1981-01-12	13.3
14	1981-01-13	16.7
15	1981-01-14	21.5
16	1981-01-15	25
17	1981-01-16	20.7

Date - Month	AVERAGE of Temprature
Jan	17.14032258
Feb	16.8
Mar	14.21774194
Apr	11.91333333
May	9.537096774
Jun	6.456666667
Jul	6.109677419
Aug	7.290566038
Sep	10.14333333
Oct	10.08709677
Nov	11.89
Dec	13.68064516

Data Transformation



Data Normalization

- Data normalization is the process of scaling the data to a much smaller range, without losing information to help minimize or exclude duplicated data and improve algorithm efficiency and data extraction performance.
- **Min-max normalization:** Where you perform a linear transformation on the original data.
- **Z-score normalization:** In z-score normalization (or zero-mean normalization) you are normalizing the value for attribute *A* using the mean and standard deviation.
- **Decimal scaling:** Where you can normalize the value of attribute *A* by moving the decimal point in the value.

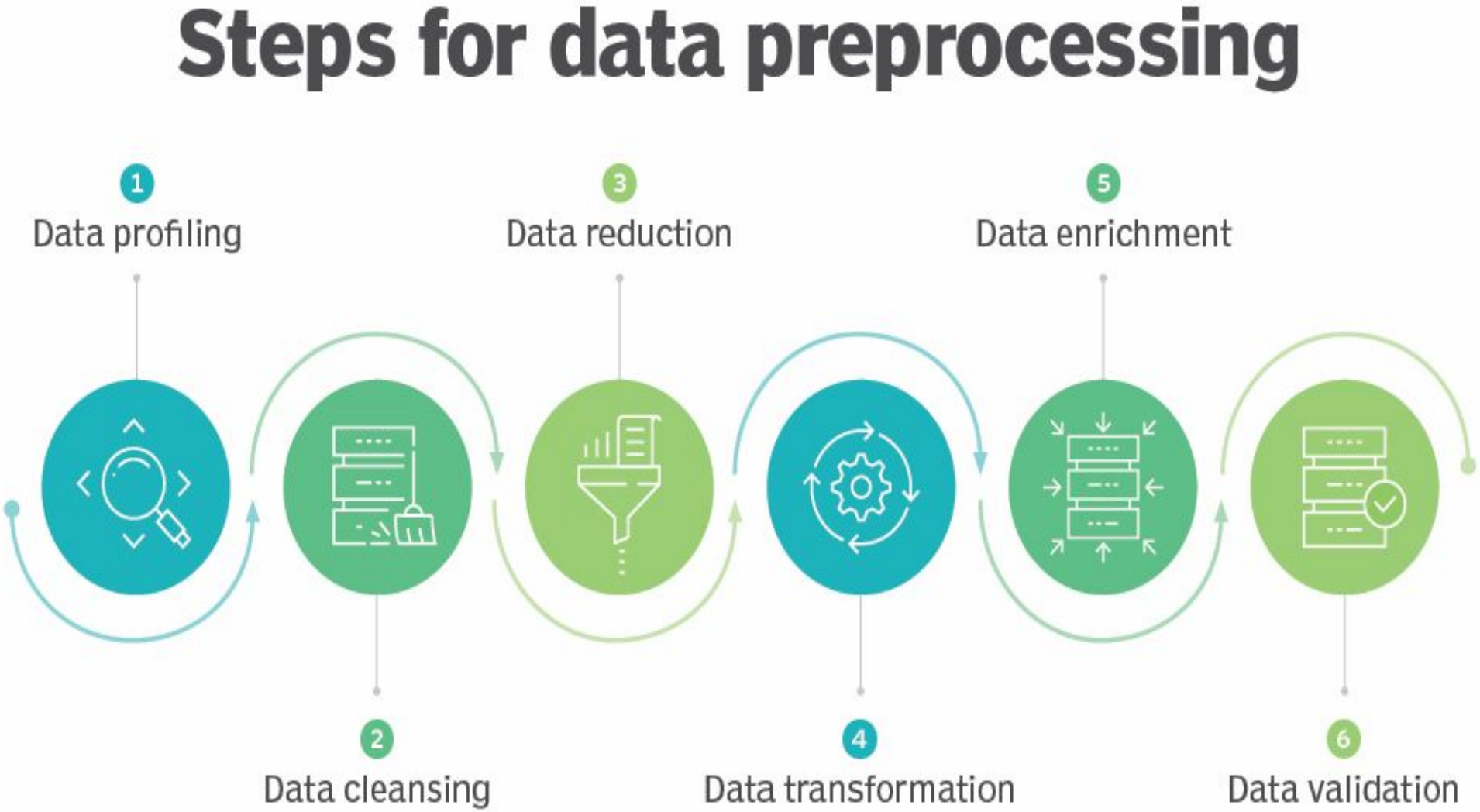
Data Transformation– some videos



- https://www.youtube.com/watch?v=kJ_f-Y6g3kw. (Tamil)

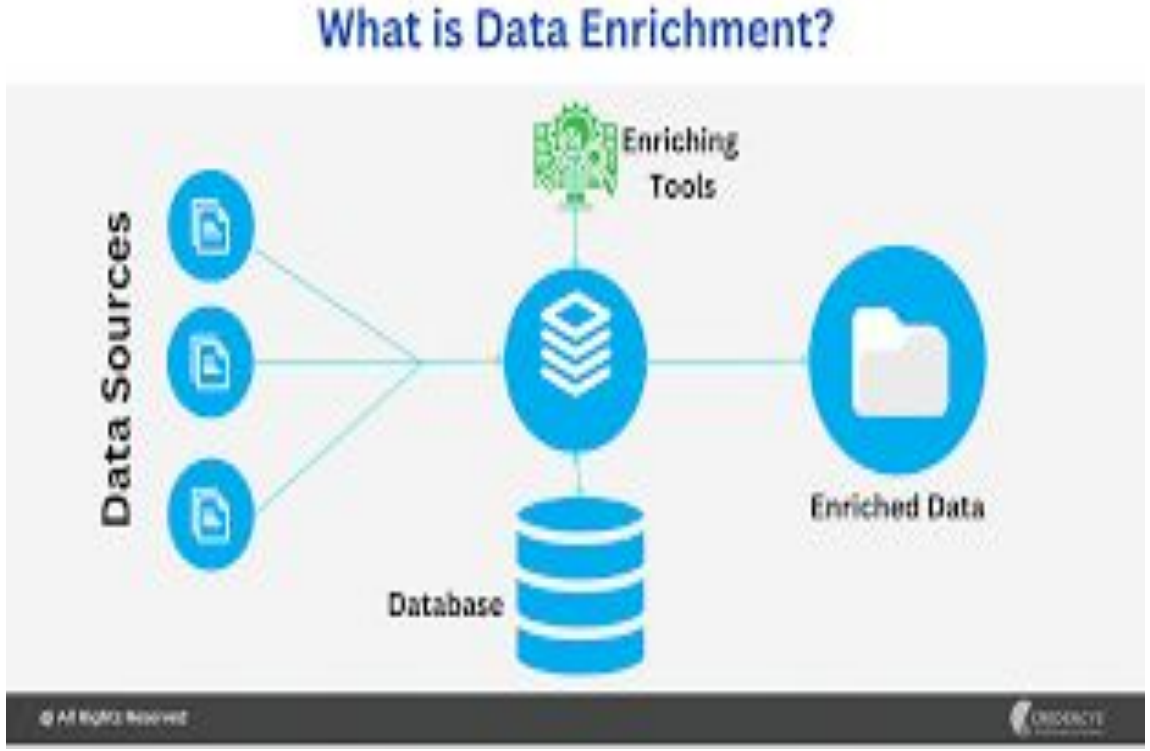
- https://www.youtube.com/watch?v=L4IYZFejj90

Key steps in Data Pre-processing



Data Enrichment

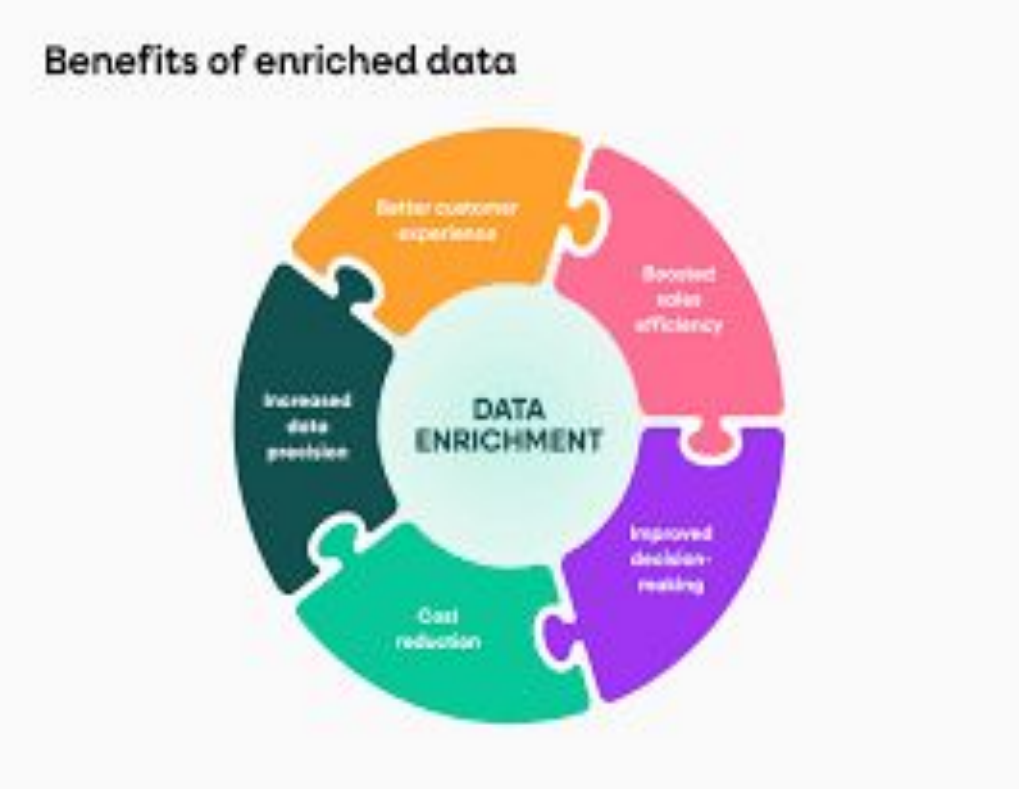
- The practice of adding more information to raw data to make it more complete and thorough is known as data enrichment.
- It includes enhancing accuracy, adding pertinent features, and closing gaps to increase the data's analytical value.
- Through this process, simple knowledge is transformed into a rich resource that may provide greater understanding.
- It may assist you in improved decision-making, process optimization, product improvement, and consumer understanding.



Data Enrichment - Importance

Data enrichment can provide many benefits for your business or organization, such as:

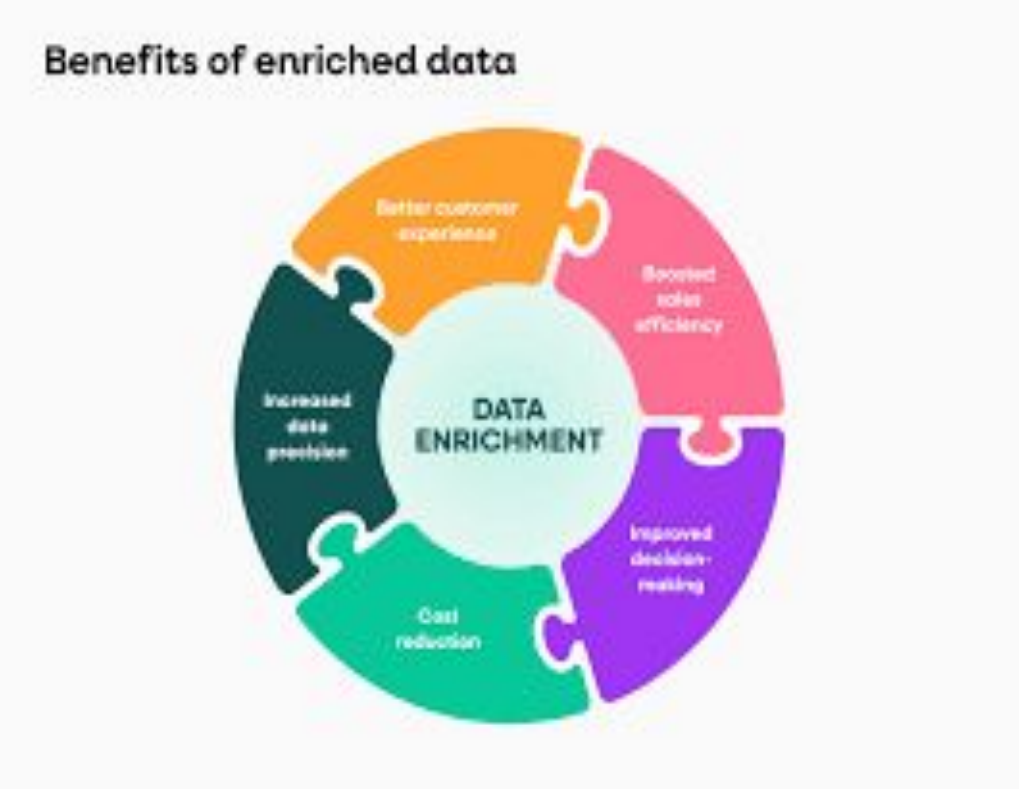
1. **Personalization:** You may better cater your goods, services, and marketing to the tastes, requirements, and actions of your target audience
2. **Customer experience:** Data enrichment can help you to enhance customer satisfaction and loyalty may be achieved through data enrichment by offering superior service and support.
3. **Better Decision-Making:** Enriched data provides a complete understanding of the subject, enabling organizations to make more informed decisions.
4. **Targeted Marketing:** Enriched data allows businesses to better understand their target audience, leading to more effective and personalized marketing strategies.
5. **Performance:** With more precise and dependable data at your disposal, data enrichment may help you enhance your operations and procedures.
6. **Improved Accuracy:** By adding missing details and correcting errors, Data Enrichment enhances the accuracy of the dataset, reducing the risk of misinformation.
7. **Innovation:** Data enrichment can help you discover new opportunities and solutions by providing you with more diverse and comprehensive data. Data enrichment, for instance, may be used to discover new markets, trends, and patterns in your data, from which you can create new features, products, and business plans.



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Data Enrichment – Implementation Steps

Depending on your data sources, objectives, and available technologies, there are several approaches to data enrichment.

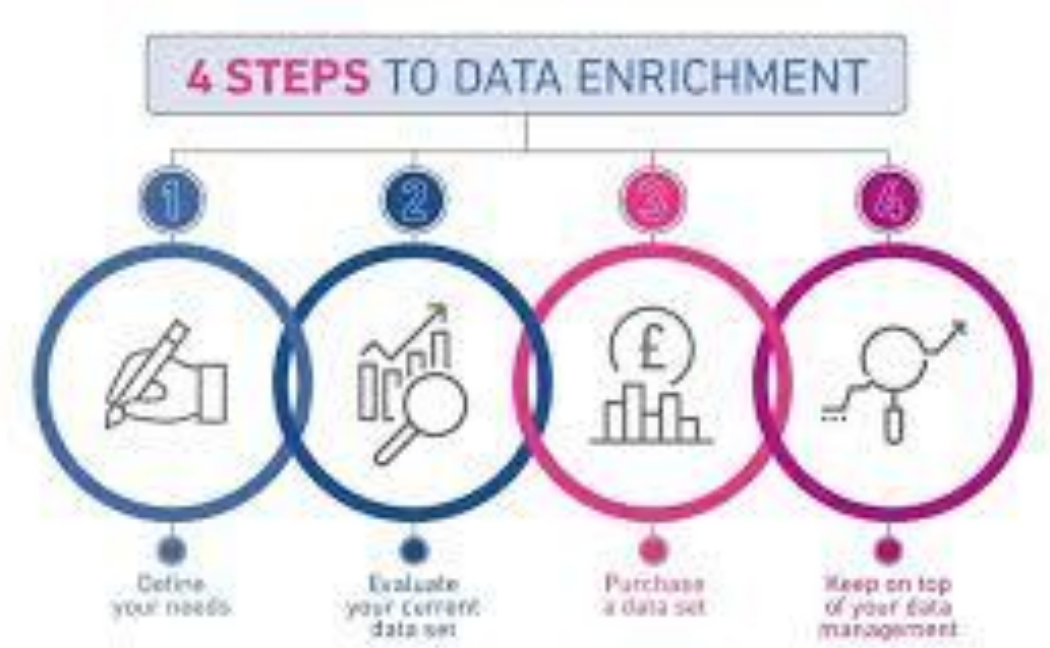
The following are some general actions to take:

Identify your data needs: You must decide what sort of information and why you wish to add it to your data before you can begin the process of enriching it.

Find your data sources: Finding the finest sources to add to your data is the next step.

Integrate your data sources: The next step is to combine your new and current data sets with your data sources.

Enrich your data: Finally, With the additional information from your data sources, you must enhance your data.



Data Enrichment – Examples



Data enrichment involves adding layers to the existing information to give it more depth and utility.

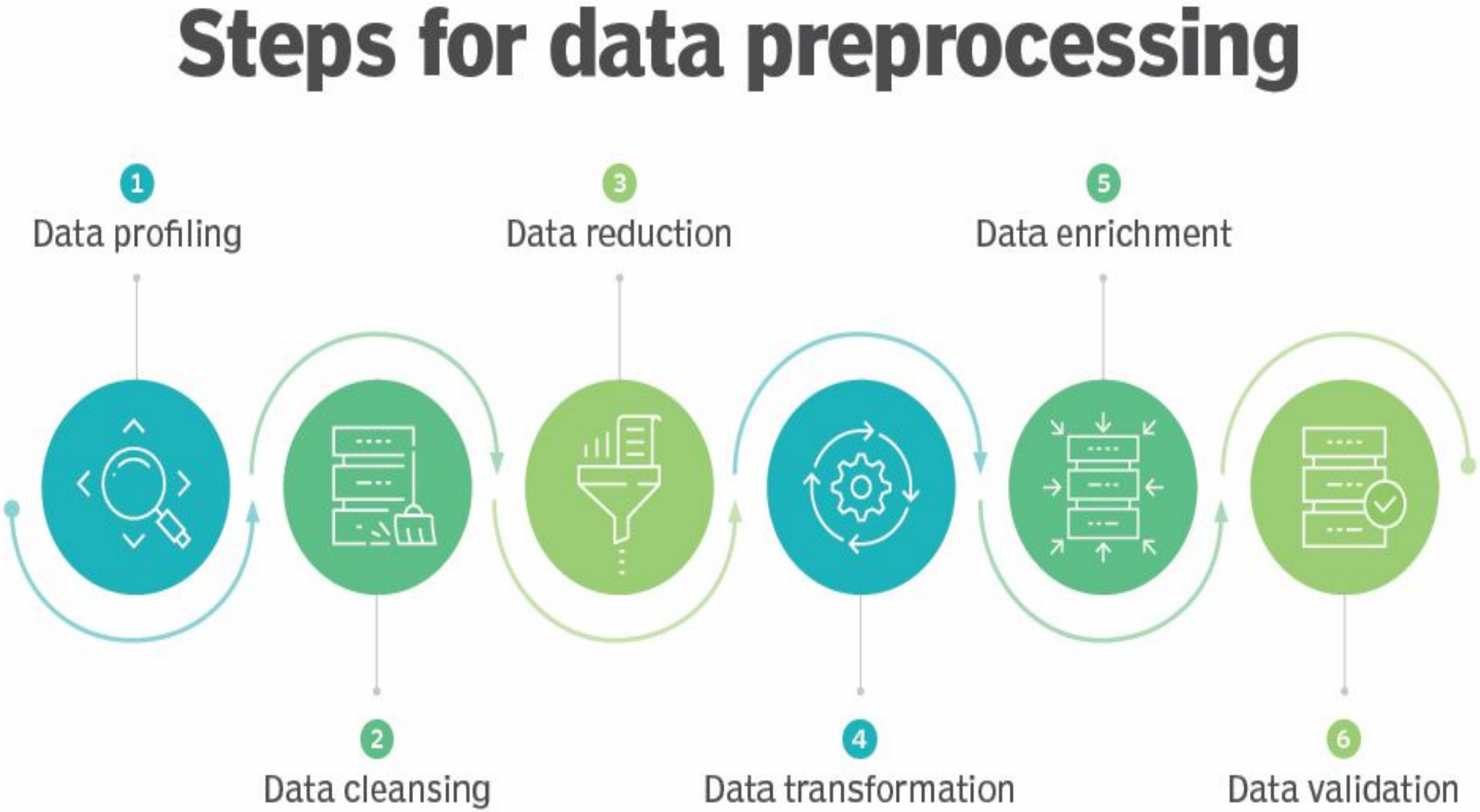
- For instance, building a contact list is significantly bolstered by data enrichment when bare-bones datasets like emails and phone numbers are augmented with details such as social media profiles or purchasing history.
- Customer segmentation is another example where customer data enrichment plays a large role for marketers who subdivide their audience into more focused groups. Basic demographic information is enriched with behavioral data, enabling more targeted and effective campaigns.
- Enriching data with industry, number of employees, and annual revenue can provide B2B sales teams with invaluable insights. They can then tailor their sales approach according to the size and industry-specific needs of the company.
- Enhancing data by integrating additional details from external sources is key to strengthening efforts against fraud. This allows companies to compare user-provided data against external databases, enabling them to more accurately flag and investigate suspect transactions.

Data Enrichment– some videos



- https://www.youtube.com/watch?v=tvphmdvSTcY
- https://www.youtube.com/watch?v=jFyTypHX1Io
- https://www.youtube.com/watch?v=pVMmSiRaqU0

Key steps in Data Pre-processing



Data Validation



- Data validation involves checking the integrity, accuracy, and structure of data before using it in business operations.
- The results of data validation can be used in data analytics, business intelligence, or to train machine learning models. It also ensures data integrity for financial accounting or regulatory compliance.
- Various methods can be used to examine data during validation, such as data type validation, constraint validation, structural validation, consistency checks, and code validation. Each type of validation ensures that the data meets the necessary requirements to be useful.
- Data validation is closely related to data quality. It is a key component in measuring data quality, ensuring that a given data set is accurate, authoritative, and of the highest quality.

Data Validation – Different Types



Multiple types of data validation are used to ensure the accuracy and suitability of data. The most common types include:

1. **Data Type Validation**: Ensures that the data in each field, column, list, range, or file matches a specified data type and format.
2. **Constraint Validation**: Checks if a data field input meets specific requirements within certain ranges, such as verifying that a data field has a minimum or maximum number of characters.
3. **Structured Validation**: Ensures that data complies with a specified format, structure, or schema.
4. **Consistency Validation**: Makes sure that data styles are consistent, such as confirming that all values are listed to two decimal points.
5. **Code Validation**: Similar to consistency checks, it confirms that codes used for different data inputs are correct, such as verifying country codes or North American Industry Classification System (NAICS) codes.

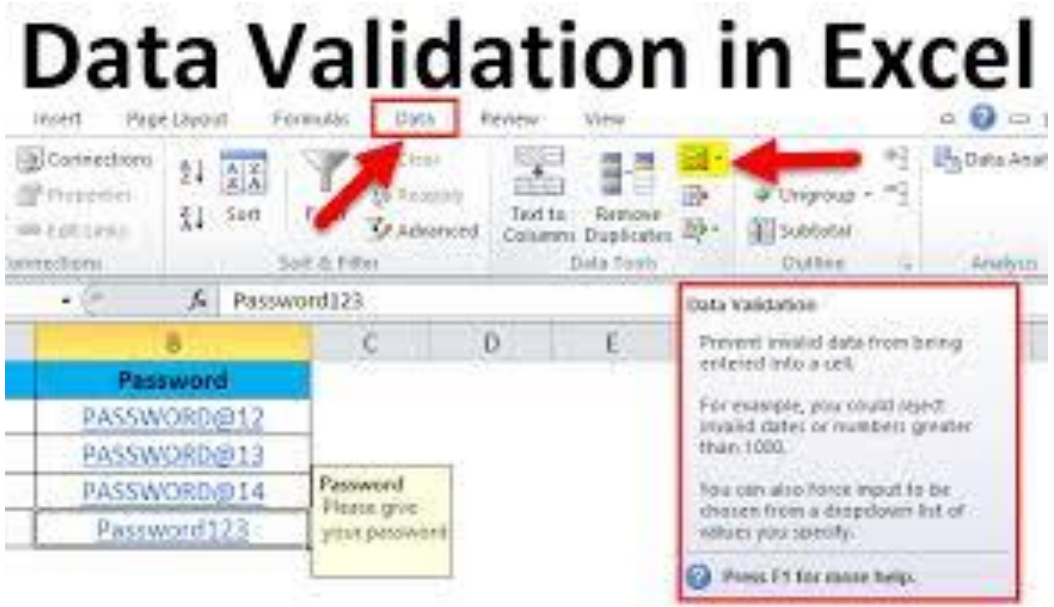
Data Validation – in X'cl

- https://www.shiksha.com/online-courses/articles/data-validation-in-excel/

ETL (Extract, Transform, and Load) and data integration tools also use data validation policies as data is extracted from one source and loaded into another.

Popular open-source tools like **dbt** include data validation options and are commonly used for data transformation.

Data validation can also be done programmatically within an application for an input value. For example, when an input variable such as a password is sent, a script can check it to ensure it meets constraint validation for the correct length.



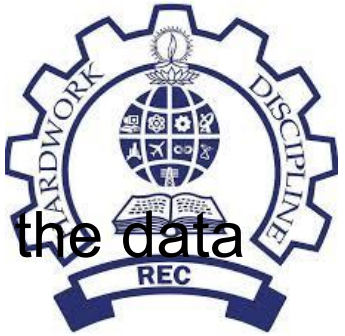
Data Validation– some videos



- https://www.youtube.com/watch?v=nMxl1_NAcxc

- https://www.youtube.com/watch?v=_g5roKHj95o

Data PreProcessing – Conclusion



- Data preprocessing is necessary before making sense of data since the data you work with might be biased and produce distorted information.

- Data preprocessing is critical in the early phases of machine learning development.

- In the AI domain, data preprocessing enhances data quality by cleaning, transforming, and formatting it to increase the accuracy of a new model while minimizing the amount of computation necessary.

- A good data preprocessing pipeline can create reusable components that make it easier to test out various ideas for streamlining business processes or improving customer satisfaction.

- Preprocessing can also simplify the work of creating and modifying data for

Attribute Subset Selection in Data Processing



Attribute subset Selection is a technique which is used for data processing .Data reduction reduces the size of data so that it can be used for analysis purposes more efficiently.

Need of Attribute Subset Selection

The data set may have a large number of attributes. But some of those attributes can be irrelevant or redundant. The goal of attribute subset selection is to find a minimum set of attributes such that dropping of those irrelevant attributes does not much affect the utility of data and the cost of data analysis could be reduced. Mining on a reduced data set also makes the discovered pattern easier to understand.

Attribute Subset Selection in Data Processing



Process of Attribute Subset Selection

The brute force approach can be very expensive in which each subset (2^n possible subsets) of the data having ‘n’ attributes can be analyzed. The best way to do the task is to use the statistical significance tests such that best (or worst) attributes can be recognized. Statistical significance test assumes that attributes are independent of one another. This is a kind of greedy approach in which a significance level is decided (statistically ideal value of significance level is 5%) and the models are tested again and again until p-value (probability value) of all attributes is less than or equal to the selected significance level. The attributes having p-value higher than significance level are discarded. This procedure is repeated again and again until all the attribute in data set has p-value less than or equal to the significance level. This gives us the reduced data set having no irrelevant attributes.

Attribute Subset Selection in Data Processing

Methods of Attribute Subset Selection

1. Stepwise Forward Selection. 2. Stepwise Backward Elimination. 3. Combination of Forward Selection and Backward Elimination. 4. Decision Tree Induction. All the above methods are greedy approaches for attribute subset selection.

Stepwise Forward Selection: This procedure start with an empty set of attributes as the minimal set. The most relevant attributes are chosen (having minimum p-value) and are added to the minimal set. In each iteration, one attribute is added to a reduced set.

In this method, we start with an empty model and iteratively add attributes that improve the model's performance.

Example:

Suppose you are building a model to predict house prices and you have these features:

Lot Size

Number of Bedrooms

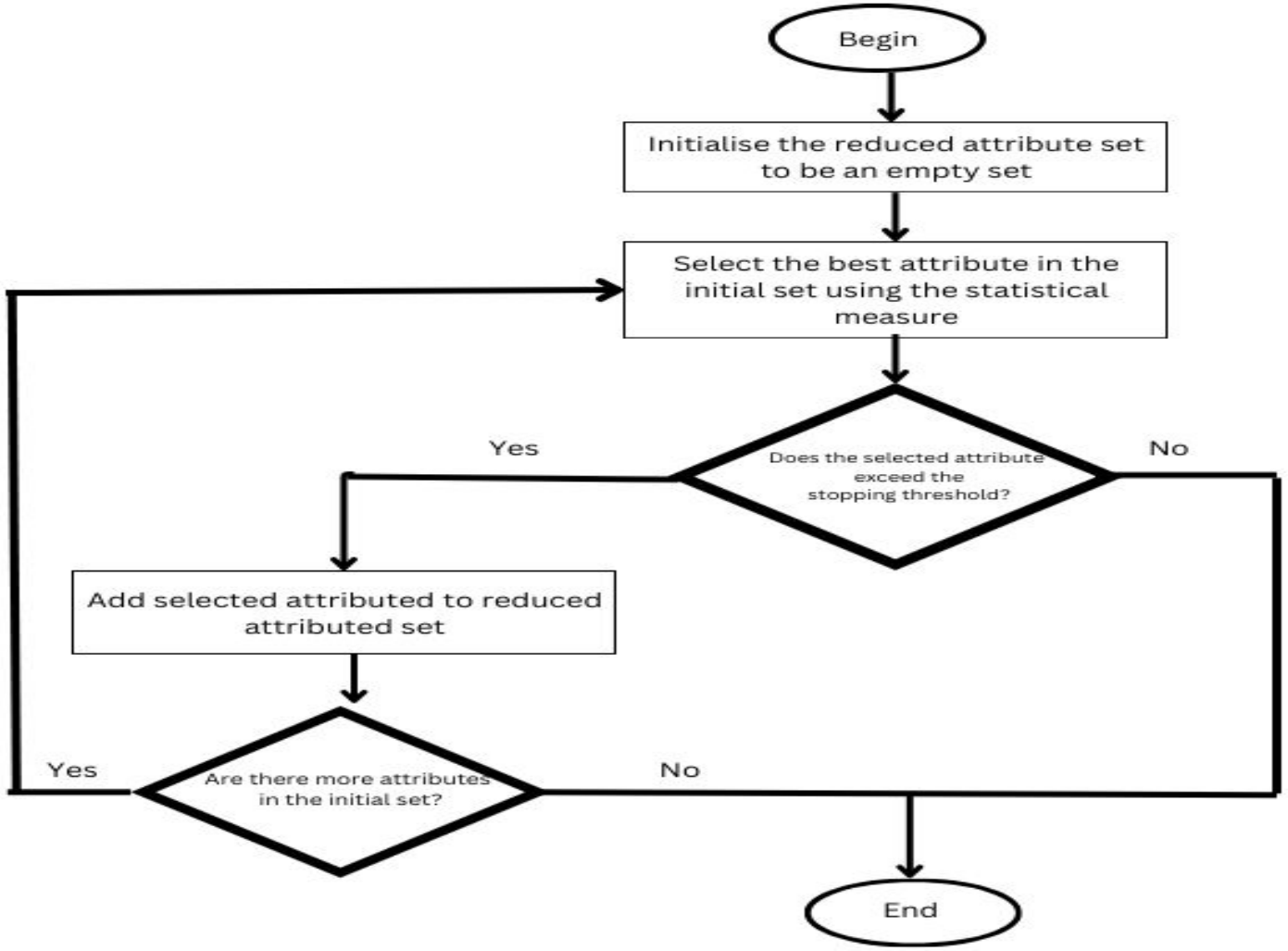
Number of Bathrooms

Garage Size

Stepwise Forward Selection:

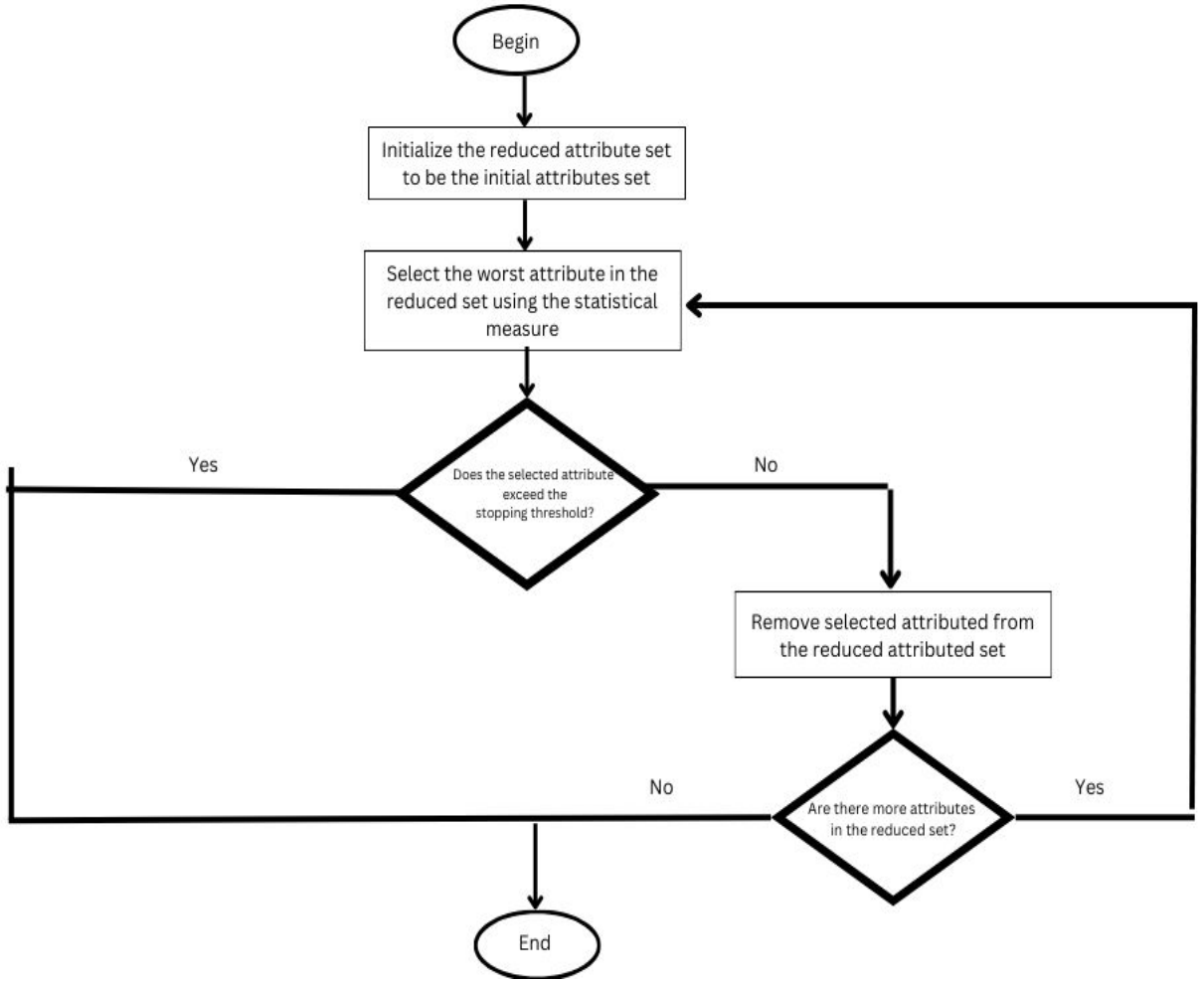
Steps:
Start: Begin with an empty model (no features).
Step 1: Evaluate model performance with each feature (Lot Size, Bedrooms, Bathrooms, Garage).
Choose the feature that gives the best improvement, say **Lot Size**.
Step 2: Add **Lot Size** and then evaluate adding each remaining feature to this set.
Adding **Garage Size** improves the model the most.
Step 3: Continue adding features (e.g., Bathrooms) until no further improvement is found.
Stop: When adding another feature doesn't improve performance.

Stepwise Forward Selection



Stepwise Backward Elimination

Here all the attributes are considered in the initial set of attributes. In each iteration, one attribute is eliminated from the set of attributes whose p-value is higher than significance level.



Stepwise Backward Elimination

Start with all features and iteratively remove the least impactful features.

Example:

Start: Begin with all features (Lot Size, Bedrooms, Bathrooms, Garage Size).

Step 1: Evaluate the performance after removing each feature:

Remove **Bedrooms** if it least affects the performance.

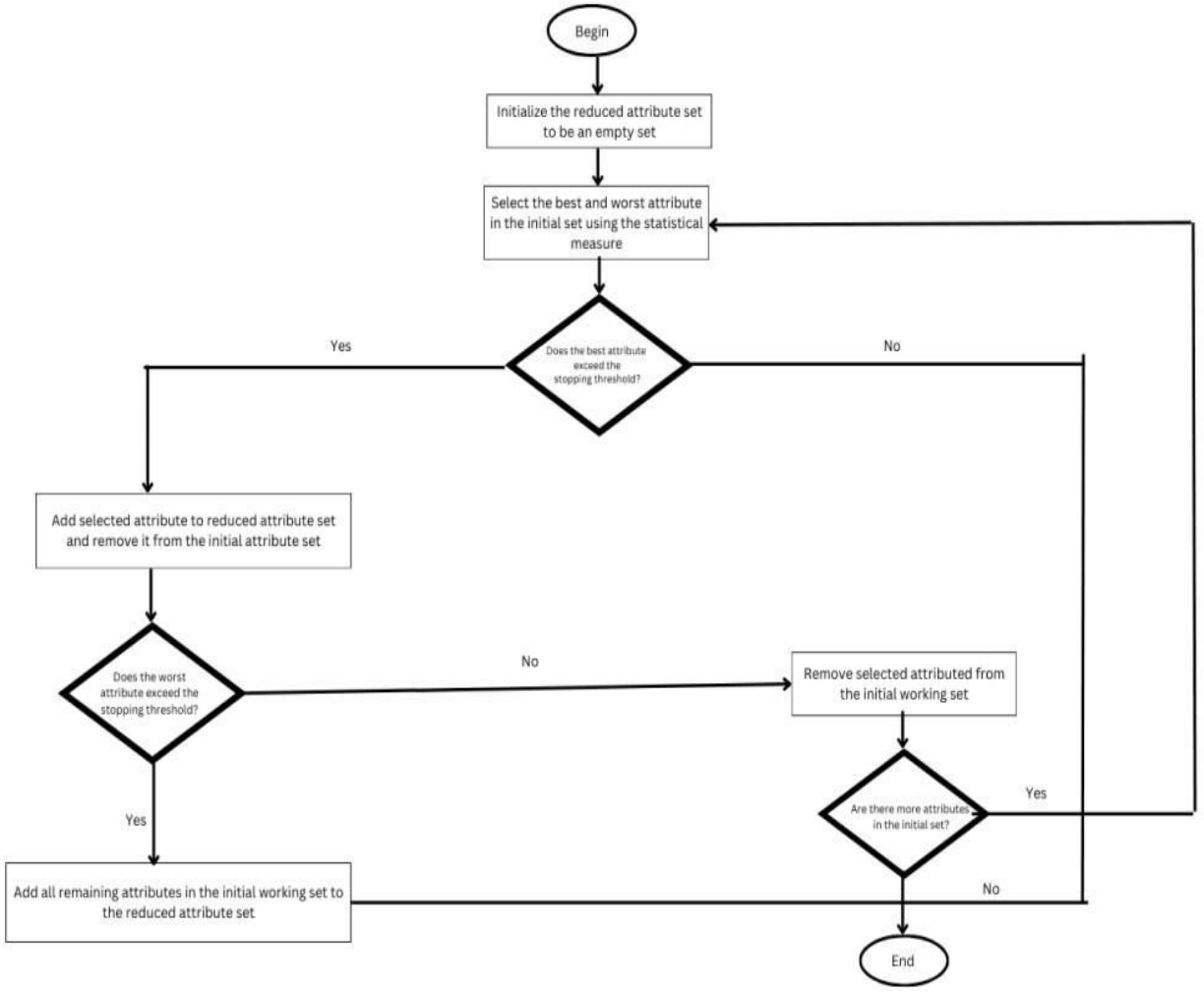
Step 2: Continue removing features, re-evaluating each time.

Step 3: Remove **Bathrooms** if performance stays the same or improves.

Stop: When removing another feature worsens the model's performance.

Combination of Forward Selection and Backward Elimination

The stepwise forward selection and backward elimination are combined so as to select the relevant attributes most efficiently. This is the most common technique which is generally used for attribute selection.



Combination of Forward Selection and Backward Elimination

A hybrid approach where you can add and remove features dynamically.

Example:

Start: Begin with an empty model or a small subset, e.g., start with **Lot Size**.

Step 1: Add the next best feature (e.g., **Garage Size**) based on model performance.

Step 2: After adding a feature, evaluate if removing any of the already added features (e.g., **Lot Size**) improves the model.

Step 3: If removing **Lot Size** improves the model, it is removed.

Stop: Continue the process until no further improvements can be made by adding or removing features.

This combination method balances both adding and pruning features dynamically to achieve the best subset of attributes for the model.

Decision Tree Induction

This approach uses decision tree for attribute selection. It constructs a flow chart like structure having nodes denoting a test on an attribute. Each branch corresponds to the outcome of test and leaf nodes is a class prediction. The attribute that is not the part of tree is considered irrelevant and hence discarded.



THANK YOU